

**Studies on variability in isolates of  
*Rhizoctonia bataticola* (Taub.) Butler causing dry  
root rot in chickpea (*Cicer arietinum* L.)**

**THESIS**

*Submitted to the*

**Jawaharlal Nehru Krishi Vishwa Vidyalaya, Jabalpur**

In partial fulfillment of the requirements for  
the Degree of

**MASTER OF SCIENCE**

*In*

**AGRICULTURE  
(PLANT PATHOLOGY)**

*By*

**Surendra Singh Patel**

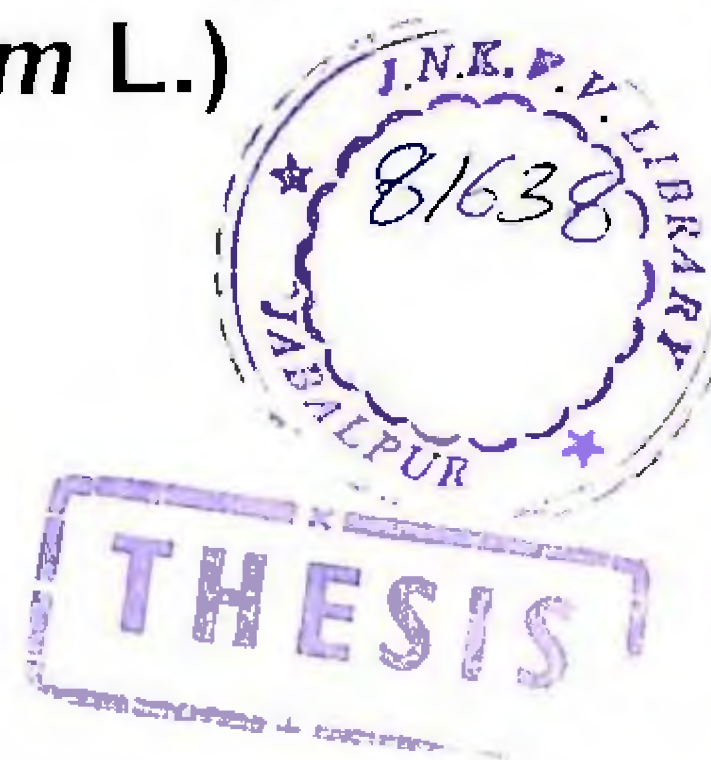
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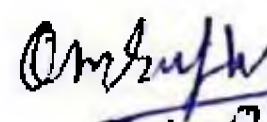
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## CERTIFICATE – I

This is to certify that the thesis entitled “**Studies on variability in isolates of *Rhizoctonia bataticola* (Taub.) Butler causing dry root rot in chickpea (*Cicer arietinum* L.)**” submitted in partial fulfillment of the requirements for the degree of **MASTER OF SCIENCE** in Plant Pathology of Jawaharlal Nehru Krishi Vishwa Vidyalaya, Jabalpur is a record of the bonafide research work carried out by **Mr. Surendra Singh Patel** under my guidance and supervision. The subject of the thesis has been approved by the Student's Advisory Committee and the Director of Instruction.

No part of the thesis has been submitted for any other degree or diploma (Certificate awarded etc.) or has been published/published part has been fully acknowledged. He has acknowledged all the assistance and help received during the course of the investigation.

  
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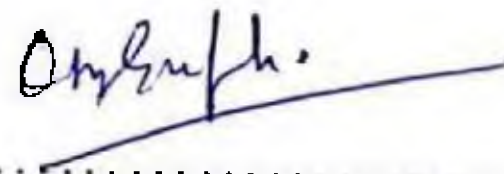
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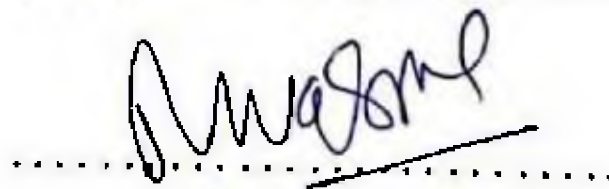
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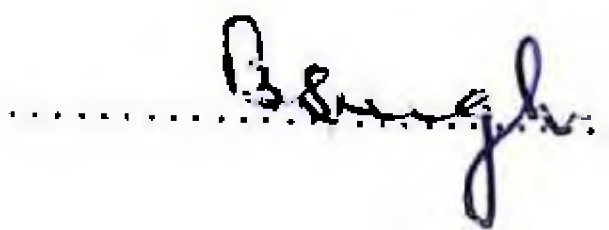
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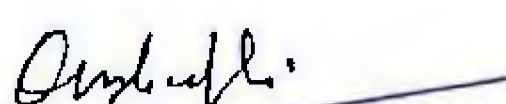
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## CERTIFICATE – II

This is to certify that the thesis entitled “**Studies on variability in isolates of *Rhizoctonia bataticola* (Taub.) Butler causing dry root rot in Chickpea (*Cicer arietinum* L.)**” submitted by **Mr. Surendra Singh Patel** to the Jawaharlal Nehru Krishi Vishwa Vidyalaya, Jabalpur, in partial fulfillment of the requirements for the degree of “**Master of Science in Agriculture**” in the Department of **Plant Pathology** has been, after evaluation, approved by the External Examiner and by The Student's Advisory Committee after an oral examination on the same.

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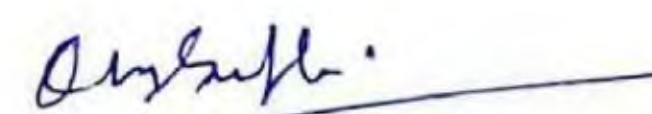
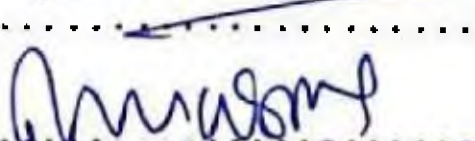
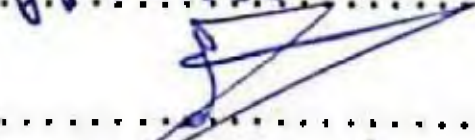
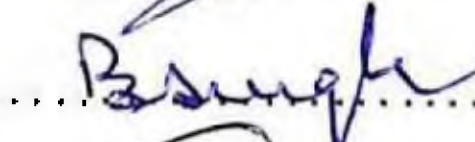
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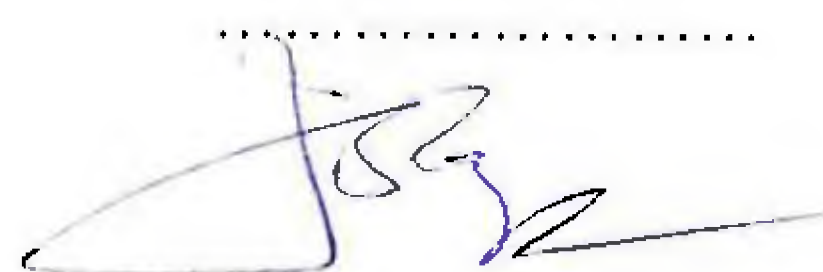
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# ABBREVIATIONS

|               |   |                           |
|---------------|---|---------------------------|
| %             | - | Per cent                  |
| @             | - | At the rate of            |
| µm            | - | Micro meter               |
| °C            | - | Degree Centigrade         |
| CD            | - | Critical difference       |
| <i>et al.</i> | - | Co workers                |
| Fig           | - | Figure                    |
| g             | - | Gram                      |
| Ha            | - | Hectors                   |
| hrs           | - | Hours                     |
| <i>i.e.</i>   | - | That is                   |
| mha           | - | Million hectare           |
| ml            | - | Milliliter                |
| mm            | - | Millimeter                |
| MR            | - | Moderately resistant      |
| mt            | - | Million tones             |
| NS            | - | Non significant           |
| psi           | - | Per square inch           |
| R             | - | Resistant                 |
| S             |   | Significant               |
| *S            | - | Susceptible               |
| SED           | - | Standard error difference |
| <i>Viz.,</i>  | - | Namely                    |
| W/w           | - | Weight/weight             |

# INTRODUCTION

In Indian Agriculture pulses play an important role not only from economical point of view but also due to their nutritional value. Chickpea is valued for their nutritive seeds with high protein content, 25.3 - 28.9 per cent after dehulling (Hulse, 1991) carbohydrate - 61.5 per cent, fat - 4.5 per cent and vitamin – 2.44. Chickpea is grown in tropical, subtropical and temperate regions. The total cultivated area of chickpea in the world is about 20.05 m.hac. and annual world production is 11.14 m. tonnes. The average yield / hac. is estimated to be around 556 kg/ha. (Ramana Rao, 2005),

The major chickpea growing countries are India, Pakistan and Turkey in Asia, Ethiopia in Africa, California and Washington state in the U.S.A., Mexico and Australia (F.A.O. 1994). In India chickpea is cultivated in 7.63 m.hac. with an annual production 5.97 m. tonnes and the average yield is estimated 782 kg / hac. (Agri. Statistics at a glance, 2006-07)

In India the major chickpea growing state are Madhya Pradesh, Rajasthan, Maharashtra and Gujarat. Madhya Pradesh contributes 46.2 percent of the total production of chickpea in the country. In Madhya Pradesh it ranks 1<sup>st</sup> in both area (2.56 m.hac.) and production (2.37 m.tonnes) and the average productivity is estimated 926 kg / hac. (Agri. Statistics at a glance, 2005-06)

Several seed, soil and air borne diseases are responsible for lowering its yield. Chickpea is infected by 67 fungal pathogens and several of them are of economic importance. Some of the diseases caused due to soil borne pathogens like *Fusarium oxysporium f.sp. ciceri* (wilt) *Rhizoctonia bataticola*. (Dry root rot) and *Sclerotium rolfsii* (collar rot), are important.

The first report of occurrence of root rot in chickpea along with wilt was made by Padwick (1948). The species *Rhizoctonia bataticola* are responsible for serious diseases (Mukharji and Bhasin, 1986) Gupta *et al.* (1983) reported incidence of root rot ranging 3.58 to 20.63 percent in 30 villages of Northern Madhya Pradesh. Shrivastava *et al.* (2002) observed highest wilt and dry root rot incidence in Sehore (25%) out of 23 locations of Vindhya plateau zone in 1998-99. This is one of the most common diseases in South East Asian countries and limits chickpea production (ICRISAT annual Report 1986: FAO, 1990).

Demiro *et al.* (1994) isolated *Rhizoctonia bataticola* from plant and seed of chickpea cultivars Azizive-94 and tested pathogenicity.

The establishment of pathotypes will be helpful in recommending varieties suitable to a particular location depending on the strain of the pathogen. Hence, the present study was undertaken with the following objectives, which will help in determining the virulence of various isolates of *Rhizoctonia bataticola* on chickpea and in identifying the host resistance.

1. To test the pathogenicity and variability in different isolates of *Rhizoctonia bataticola* collected from different districts of Madhya Pradesh, Chhattisgarh, Maharashtra and Gujarat
  - (1) Cultural
  - (2) Morphological
  - (3) Pathological (virulence)
2. Identification of host resistance under multiple disease sick field condition

# REVIEW OF LITERATURE

## 2.1 Historical

The first mention of a plant disease, which may be referred with, certainly to *Rhizoctonia* as a causal agent is an important paper by Du Hamel (1728) read before the Paris Academy. In this paper he gave a careful description of a fungus disease of saffron (*Crocus sativus*) occurring in France. He named the fungus *Tuberorides*.

De Candolle (1815) discovered a similar disease on alfalfa. In his taxonomic description he was first person who gave the name for the fungus. Later, after careful study of the characteristics and parasitism of the species, he established (1815) the genus *Rhizoctonia*.

In 1924, Small described a *Rhizoctonia*, which develop black sclerotia of variable size and shape in the diseased roots, many woody plants from Uganda and he named it *Rhizoctonia lamrllifera*. Butler (1925), however, considered that the fungus should be regarded as *Rhizoctonia* and made the combination *Rhizoctonia bataticola* (Taub.) Butler to a *sclerotium* causing a root rot of cotton, french beans and cowpeas.

## 2.2 Taxonomy

The most important taxonomic work on the genus *Rhizoctonia* in the 19<sup>th</sup> century is that of Tulasne brothers (1862). They published the first extensive morphological study on the genus *Rhizoctonia*. They reduced *Rhizoctonia crocorum* DC. and *Rhizoctonia medicaginis* DC. to a single species to which they applied a new name *Rhizoctonia vivolaceae* Tul. This reduction to a single form was made after a most careful morphological study of the fungus at all stages.

Among the taxonomic and pathological studies on *Rhizoctonia* in India, the most important is that of Shaw. Before 1912, he had studied two species of *Rhizoctonia*, one of which was characterized by its formation of small black sclerotia. The systematic position of *Rhizoctonia bataticola* as follows (Alexopoulos and Mims 1983).

|              |   |                     |
|--------------|---|---------------------|
| Kingdom      | - | Mycetae             |
| Division     | - | Amastigomycota      |
| Sub-division | - | Deuteromycotina     |
| Class        | - | Deuteromycetes      |
| Sub-class    | - | Coelomonycetidae    |
| Order        | - | Spharopsidales      |
| Family       | - | Spharopsidaceae     |
| Genus        | - | <i>Macrophomina</i> |
| Species      | - | <i>phaseolina</i>   |

## 2.3 Pathogen

The species description of *R. bataticola* as follows (Subramanian, 1971)). "The basic characteristics of the species are the formation of sclerotia in the host tissue as well as in culture. On the infected plants sclerotia seen in the pith region and outer surface of roots just below the bark region. They are brown or jet black, minute, smooth externally, hard, variable in shape: globose, oval, oblong, elliptical, curve or even forked and varying in size, but of uniform texture, more or less loosely packed."

The cell of hyphae are barrel shaped, branching more or less at right angles and light to dark brown in colour, composed of free thick-walled cells. The species *Rhizoctonia* is distinguished by the light brown-to-brown hyphae with somewhat loose/spongy sclerotia in the form of bulbils. According to Mehrotra

and Aneja (1990) both the species *R. bataticola* and *R. solani* have the variable capacity to produce enzymes.

Jharia (1970) also found that *Rhizoctonia bataticola* is associated with seeds of urid both internally and externally. Thus it seems that *Rhizoctonia bataticola* acts as both soil and seed borne pathogen from Madhya Pradesh.

## 2.4 Symptoms

*Rhizoctonia bataticola* is a soil borne fungus/pathogen. Several workers have reported various types of symptoms produced on plants with varying intensity of the disease in different crops and different regions. Narsimhan (1929) observed that plants infected with *R. bataticola* initially showed drooping of leaves followed by wilting. The main roots of such plants show blackening of finer roots.

According to Mehrotra and Aneja (1990) the first symptom of the disease is yellowing of the leaves within a day or two such leaves drop and in the course of the next two or three days whole plant dies off. The leaves and stem of the affected plants are usually straw coloured. The taproot is dark and quite brittle in dry soil.

Singh *et al.* (1990) observed the roots inoculated with *R. bataticola* [*Macrophomina phaseolina*] and reported disintegration of cortical tissues while mycelium and sclerotial bodies plugging the xylem were seen in longitudinal sections. The sudden collapse of infected plants at the pod-filling stage may be due to this plugging of the xylem vessels.

## 2.5 Survey and Surveillance

Singh *et al.* (2002) conducted a survey in Vindhyan plateau zone of Madhya Pradesh, India to evaluate the prevalence of wilt and dry root

rot disease in chickpea. They reported that *Fusarium oxysporum* was most prevalent (56 per cent) in wilted plants followed by *Rhizoctonia bataticola* (17 per cent) and *Trichoderma* sp. (10 per cent). The incidence of *Rhizoctonia bataticola* also caused withering and drying of plants, which ranged from 8 to 20 per cent.

Reddy *et al.* (1991) stated that the countries like Mandalay, Sayaing and Bago, central and lower Myanmar the major disease in chickpea was dry root rot caused by *R. bataticola*. Aung *et al.* (1992) observed that dry root rot caused by *R. bataticola* was major disease of chickpea.

## 2.6 Occurrence and losses

The first report of the occurrence of root rot in chickpea alongwith wilt was made by Padwick (1948). The incidence of root rot ranging from 3.58 to 20.63 per cent in 30 villages of northern Madhya Pradesh was reported by Gupta *et al.* (1983). This is one of the most common diseases in south East Asian countries and limits chickpea production (ICRICAT Annual Report 1986; FAO, 1990).

Tripathi and Sharma (1983) studied the incidence of chickpea dry root rot (*Rhizoctonia bataticola*) in Southern Haryana and reported that the incidence was high from late October to mid November which was described in December and January and then again increased in the months of February and March. They also noted that seed sown at a depth of 15 cm developed a less root rot then that sown at 7.5 cm.

The *Rhizoctonia bataticola* responsible for serious diseases like stem blight, foliage blight, root rot, seedling blight, charcoal rot, wilt, collar rot, dry rot, pod and seed rot (Raut and Ingul, 1989) in several crops such as soybean, gram mung, urid, fruit trees and many vegetables (Mukharji and Bhasin, 1986). This

sclerotial fungus is of wide geographical distribution ranging from Formosa, Philippines, India, Ceylon, East Africa, Palestine and Egypt to the West Indies and the Eastern United State (Ashby, 1927). Significant reduction (25-70%) in chickpea production is caused by this disease in India (Ahmed *et al.* 1986 and Pandey and Singh, 1990).

## 2.7 Pathogenicity

Nema and Khare (1973) reported that chickpea cultivation in Madhya Pradesh was greatly threatened by soil borne diseases generally appearing at two phases of growth, early seedling stage and flowering stage, wilting was due to *Fusarium oxysporum* f. sp. *ciceri*, dry root rot caused by *Rhizoctonia bataticola* and *Sclerotium rolfsii* resulted in collar rot.

Dermiro *et al.* (1994) observed that dry root rot disease of gram is also becoming a serious disease affecting productivity of this crop. They isolated *Rhizoctonia bataticola* from plants and seeds of chickpea cultivars Azizive 94 and tested pathogenicity.

## 2.8 Variability

### 2.8.1 Cultural Variability

Meenakshi Shekhar *et al.* (2006) studied seven isolates of *Macrophomina phaseolina* a causative agent of charcoal rot of maize, obtained from different agro-ecological zones of India, and found that they varied in their cultural characteristics and pathogenic behavior. On the basis of colony colour, isolates were divided into four groups i.e., grayish white, blackish gray, dark black in center periphery creamish and cottony white color.

Byadgi and Hegde (1985) studied the variation in isolates of *R. batitocola* obtained from six different plant species differed in their cultural characters, growth rate, morphology of the sclerotia, pycnidia and pycnidospores and virulence. Cultural characters of the isolates after 3 days of incubation indicated bean isolate grows significantly faster followed by Bengal gram and cowpea isolates, which produced fluffy dark brown mycelium. The sorghum and soybean isolates did not differed significantly and exhibited moderate growth with the partially fluffy mycelium. The gliricidia isolate produced poor growth and appressed light brown mycelium. The colonies had smooth margins in all the cases.

Jharia (1985) isolated *Rhizoctonia bataticola* from three different sources. He observed variation in their growth rate. The soil isolate was fast growing (90mm) followed by root (89mm) and seed (67mm) isolates on potato dextrose agar medium. Sobti and Bansal (1988) studied variation in cultural characteristics of root, leaf and seed isolates of *Rhizoctonia bataticola in-vitro* tests on different media the seed and root isolates had the smallest and large sclerotia respectively. Growth characteristics of potato dextrose agar, Richard's medium and groundnut kernel agar is described.

According to Ratnoo *et al* (1997) variation in imperfect fungi is not uncommon; they differ in cultural, morphological and pathogenic characters. They studied two isolates of *M. phaseolina* from cowpea roots and stem showed variability in their cultural characteristics, from which it was found that M1 isolate showed dark grey mycelium, aerial fluffy growth, hyphae thin become thicker with age, septate and profusely branched. Sclerotial formation abundant, blackish, globose to round and measured 49-76  $\mu\text{m}$  in diameter whereas M-2 isolate exhibited grey, aerial cottony, compact regular and fast growth of mycelium, hyphae thin, septate branching to form bunch like structure.

Hooda and Grover (1982) studied fourteen isolates of *Macrophomina phaseolina* obtained from different plant species and plant parts of the same host

differed in their culture characteristics and pathogenic behaviour, the isolates exhibited wide variation in mycelial dry weight as well as sclerotial production. Among the thirteen isolates from four leguminous crops, least growth was noted in isolates from mungbean stem, mungbean pods and cowpea leaf, whereas mungbean leaf and seed isolates showed significantly higher growth than others. Mungbean seed isolate had very fluffy growth. Castor isolate had significantly higher mycelial dry weight than other isolates, but produced few sclerotia. There was no correlation between the sclerotial production and mycelial dry weight among the isolates obtained from the same host tissues.

Monga and Sheo Raj (1994) studied thirteen isolates of *R. solani* and categorized into four distinct groups on the basis of cultural characteristics. The cultural characteristics of different isolates revealed that isolate in group Rs-2 was faster in mycelial growth which was significantly more than any other isolate belonging to other groups. However, isolates in Rs-1 and Rs-3 did not show significant growth differences among themselves.

Jain *et al.* (1973) found *Rhizoctonia bataticola* associated with root, stem, leaf, pod and seeds of *Phaseolus mungo*. The isolates from various plant parts and soil showed differences in virulence. The soil isolate was most pathogenic. The isolates differed in their growth pattern and sclerotial size. The leaf isolate developed the largest sclerotia and the seed and soil isolates the smallest. When grown on different media the isolates varied in growth pattern and growth rate.

### **Morphological Variability**

Meena Shekhar *et al.* (2006) reported variation in morphological characteristics of 7 isolates of *Macrophomina phaseolina*. On the basis of sclerotial morphology, two groups of isolates could be formed, the one with oblong shape having irregular edges and the other being round with regular edges. On the basis of

number of sclerotia Hyderabad isolate produced highest number of sclerotia 52.0 where as Coimbatore isolate produced minimum number of sclerotia 48.8 / 10x microscopic field

Byadgi and Hegde (1985) studied the variation in Isolates of *R. bataticola* obtained from six different plant species differed in their cultural characters, growth rate, morphology of the sclerotia. Gliricidia isolate produced the biggest sclerotia with mean diameter of 101.51  $\mu\text{m}$ , while cowpea isolates produced the smallest sclerotia (66.88  $\mu\text{m}$ ). The sclerotia at maturity were of various shapes varying from globose to irregular and brown to black in colour, appearing hairy owing to the dark hyphae emerging from them.

Mandal *et al.* (1998) studies the sclerotial morphology and pathological reactions of 13 isolates of *M. phaseolina* from jute growing regions of India. He found that sclerotia size ranged from 66.14 to 128.25  $\mu\text{m}$  and the colour varied from light to dark brown. Sclerotial shape in most cases was irregular except for few which were round to elongate.

Dhingra and Sinclair (1972) studied 15 isolates of *M. phaseolina* and reported morphological variations in isolates from root, pod and seed of soybean plants. They observed that sclerotia from root isolates ranged from 122 – 188 x 77 – 127  $\mu\text{m}$ , while stem isolates 84 – 100 x 60 – 71  $\mu\text{m}$ .

Critopoulos (1953) compared two isolates of *R. bataticola* and reported variation in their morphological characteristics he found that in one isolate, sclerotia were spherical to ellipsoid 60-165 X 57-114  $\mu\text{m}$ , while in the others round to oval measuring 36-99 X 36-81  $\mu\text{m}$ .

Sobti and Sharma (1992) studied 13 isolates of *Rhizoctonia bataticola* of groundnut collected from Rajasthan. Cultural, morphological characteristics and

virulence of the isolates were investigated and reported that hyphal cell size ranged from 15.30-24.64 X 4.16-8.48  $\mu\text{m}$  and sclerotial size ranged from 73.88-188.33 X 51.09-155.90  $\mu\text{m}$  and the colony diameter of the isolates ranged from 59.00 to 84.00 mm.

According to Raut and Ingle (1989) the isolates of *Rhizoctonia bataticola* from different cultivated crops belonging to seven families varied in their growth rate in terms of size of sclerotia, production of pycnidia and size of pycnidiospores. Differences were evident even between the isolates of crop belonging to same family. Light was found essential for pycnidia formation. Comparatively cotton and cowpea isolates formed more sclerotia. The size of sclerotia varied with the host. The sclerotia of cotton isolate were the largest (mean dia. 115.06  $\mu\text{m}$ ) and soybean isolates the smallest (56.33  $\mu\text{m}$ ). Cluster bean, pigeonpea and black gram isolates measured between 61 - 70  $\mu\text{m}$ , safflower, okra, green gram, chilli, sunflower and citrus isolates between 71 - 80  $\mu\text{m}$  and cow pea, sorghum, chickpea and sesamum isolates between 81 - 90  $\mu\text{m}$ .

Monga and Sheo Raj (1994) obtained seven cultures of *R. bataticola* during isolation from cotton root rot samples based on morphological characteristics, sclerotial number and size; it was observed that these cultures were distinct isolates of *R. bataticola*. For instance the size of sclerotia varied from 58.83 $\mu\text{m}$  to 126.63 $\mu\text{m}$ . similarly, these were notable difference in sclerotial number and colony characteristics also. Isolate 1 produced minimum number of sclerotia, where as isolate 3 and 6 produced abundant sclerotia. Growth studies revealed that isolate 1 had significantly lesser growth as compared to other isolates. Isolate 3 shows significantly higher growth compared to other isolates. Incidentally, these isolates also produced abundant large size sclerotia.

## 2.8.2 Pathological Variability

Meena Shekhar *et al.* (2006) reported variation in pathogenic behavior of 7 isolates of *Macrophomina phaseolina*. On the basis of disease expression the Hyderabad isolate was observed to be most virulent by giving maximum disease (8.8) while Coimbatore isolate were found to be least virulent as it exhibited a highest rating of 5.2 in the susceptible inbred CM 120 on a scale of 1 to 9.

Byadgi and Hegde (1985) studied the variation in Isolates of *R. batitocola* obtained from six different plant species which also differed in virulence. Based on the variation in virulence, the isolates obtained from six different host plants were classified into three groups, A (Virulent) B (Intermediate) and C (Mild).

Ratnoo *et al.* (1997) studied two isolates of *Macrophomina phaseolina* established from cowpea root and stem showed variability in cultural and pathogenic characters. The isolate M<sub>2</sub> proved to be more virulence with disease index 86.11% compared to isolate M<sub>1</sub> (77.7%). Isolate M<sub>2</sub> produced the symptoms in 7-10 days and blighted the plants in 10-15 days compared to delayed production of symptoms (9-14days) and blighting (25-35days) by isolate M<sub>1</sub>.

Singh *et al.* (1990) studied five different isolates of *R. solani* on the basis of pathogenic behavior. All isolates were pathogenic on all the five varieties tested but they differed in their degree of virulence. The isolates A and B were more virulent than the isolates C, D and E. Isolate E was found to be the least virulent among all the isolates tested.

Sobti and Sharma (1992) during 1984-85, collected 13-groundnut samples infected by *R. bataticola* [*Macrophomina phaseolina*] in Rajasthan, India. The pathogen was isolated, purified and named according to the location

at which it was collected and virulence of the isolates were investigated. The disease intensity varied from 13 – 16 per cent. The Tonk isolates were highly virulent whereas CHG and BHW II had the lowest virulence. No correlation between radial growth and virulence or virulence and pycnidial formation was found.

Monga and Sheo Raj (1994) tested two isolates for pathogenicity at 3 inoculum levels. Isolate 7 was more pathogenic than isolate 1. Increase in inoculum level of 2 to 4 per cent significantly increased seedling mortality.

## 2.9 Host resistance

Nene *et al.* (1989) identified broad based stable resistant lines to wilt and root rots on the basis of multilocation evaluation, (ICC 11550, ICC 9032, ICC 9023, ICC 2862 and ICC 11551).

Reddy *et al.* (1990) reported two chickpea varieties *i.e.* Avrodhi and ICC 98 resistant to wilt and root rots from ICRISAT during 1985-86 to 1988-89.

Bhatt (1993) observed that BGM 416, BG 416, ICC 1376 and IC 11314 were resistant to root rot. Resistant cultivars had greater number of total roots during early growth phases.

Gupta (1997) identified genotypes ICCV 89402, ICCV 90201, ICCV 90254, BG 209, H 88-2, GF 88426 and RSG 180 having dual resistance against wilt and dry root rot.

Gangwar *et al.* (2002) screened 35 chickpea cultivars for resistance to dry root rot caused by *Rhizoctonia bataticola* in a field experiment. All the cultivars exhibited disease symptoms, only 10 cultivars (ICC 2644, 10384, 10630, 112244, 11332, ICCL 81002, 810810,

ICC 12263, 12441 and ICCV 90254) were resistant whereas 5 genotypes (GCP 9504, Phule G 96020, 96105, 96313 and GL 91059) were moderately resistant to dry root rot.

Prajapati *et al.* (2003) while working on resistant sources of chickpea against dry root rot (*Rhizoctonia bataticola*) evaluated forty-five chickpea cultivars for resistance to dry root rot. Among these, H 355, JG 315, Pusa 212, G 543, ICC 32, BG 244, Phule G 5, ICC numbers 2644, 10384, 11224, 10630 and 11332, ICCV 10, ICC numbers 12441 and 12263, 81010 and 81002, Avrodhi, ILC 702 and ILC 200 cultivars were rated as resistant. Five cultivars were moderately resistant and 9 were tolerant.

Pandey *et al.* (2004) evaluated twenty-nine chickpea germplasm accessions, 10 cultivars and 8 advanced breeding lines for resistance to dry root rot, caused by *Rhizoctonia bataticola* under *in vitro* conditions. They observed that one germplasm accession (ICC 14395), a cultivar (ICCV 2) and advanced breeding line were resistant to dry root rot. Of the remaining lines 22 were moderately resistant, 19 susceptible and 2 highly susceptible line (BG 212 and ICC 12267) used as the control.

Gupta and Babbar (2006) conducted a field experiment to search the multiple as well as individual disease resistant sources against various soil borne diseases in chickpea. The results revealed that the pathogens *Sclerotium rolfsii* and *R. bataticola* caused severe mortality ranging from 1-74 and 1-100 percent at seedling and later stages of crop growth. However, the genotypes, which performed well, exhibited <10 per cent mortality against these diseases.

# MATERIALS AND METHODS

Present investigations were undertaken to study the "variability in isolates of *Rhizoctonia bataticola* (Taub.) Butler causing dry root rot in chickpea (*Cicer arietinum* L.)." The materials used and methods followed are described below;

## A. MATERIAL

### 1. General

Borosil make glassware's were used throughout the investigation. These were cleaned with cleaning solution followed by through washings with detergent powder and subsequent rinsing with water. Glassware's were sterilized in hot air oven at 180<sup>0</sup> C for two hours.

The inoculation needle and other metallic instruments were sterilized by dipping them in an alcohol and heating over flame. The culture media were sterilized in an autoclave at 1.05 kg pressure per square inch. (15 lbs., p.s.i / at 121.6<sup>0</sup>C) for 15 minutes. At the time of pouring of plates, streptomycin @ 600 mg/l was added to the medium to inhibit the bacterial growth. The soil/ sand and chickpea straw was sterilized at 2.1kg pressure per square inch (30 lbs. pressure p.s.i) for two hours.

### 2. Test pathogen

*Rhizoctonia bataticola* the causal organism of chickpea dry root rot.

### 3. Culture Media

The following media were used during the course of investigation.

#### (a) Potato dextrose agar (P.D.A)

|                 |   |         |
|-----------------|---|---------|
| Peeled potato   | - | 200 g   |
| Dextrose        | - | 20.0 g  |
| Agar            | - | 20.0 g  |
| Distilled water | - | 1000 ml |

(b) Potato dextrose broth (P.D.B.)

|                 |   |         |
|-----------------|---|---------|
| Peeled potato   | - | 200 g   |
| Dextrose        | - | 20.0 g  |
| Distilled water | - | 1000 ml |

For potato extraction peeled and sliced potato was taken and boiled in 500 ml water. Than it was strained through muslin cloth. Agar- agar was washed with water and taken in 500ml of water then heated until it melted fully. Melted agar agar was poured in potato extract and dextrose was added and the solution was finally made-up to one liter. The media was poured in conical flask and sterilized in an autoclave.

In PD broth similar procedure as described above was employed only agar agar was not added in potato dextrose broth.

## **(B) METHODS**

### **1. Collection of diseased plant / root sample**

Chickpea plants showing symptoms of dry root rot were procured from the wilt net work project (chickpea) collected at flowering and pod formation stage from the 40 locations of 23 districts surveyed in four major chickpea growing stases viz; Madhya Pradesh, Chhattishgarh, Maharastra and Gujarat. Samples were brought in the laboratory and symptoms were recorded in detail. The tissue of affected plant showing the symptoms of dry root rots were examined under the microscope for the presence of the causal agent. The different locations of collection are given in the Table. - 1.

Table. - 01 (A): Collection of diseased plant / root samples of chickpea, collected from different districts of Madhya Pradesh, Chhattisgarh, Maharashtra and Gujarat

| States         | Isolates no. | Isolate Code. | Districts  | Village/Locations |
|----------------|--------------|---------------|------------|-------------------|
| Madhya Pradesh | 1            | JBP1          | Jabalpur   | Breeding Farm     |
|                | 2            | D2            | Dindori    | Shahpura          |
|                | 3            | D4            | Dindori    | Amachuha          |
|                | 4            | D6            | Dindori    | Karanni           |
|                | 5            | P3            | Panna      | Phulwari          |
|                | 6            | BT4           | Betul      | Kolgoan           |
|                | 7            | RAI4          | Raisen     | Katia             |
|                | 8            | RAI5          | Raisen     | Rangai            |
|                | 9            | K2            | Katni      | Maiher            |
| Chhattisgarh   | 10           | RA2           | Raigarh    | Chandrapur        |
|                | 11           | BIL1          | Bilaspur   | Saragoan          |
|                | 12           | BIL2          | Bilaspur   | Pathariya         |
|                | 13           | BIL3          | Bilaspur   | Baitalpur         |
|                | 14           | BIL4          | Bilaspur   | Bhatapara         |
|                | 15           | BIL5          | Bilaspur   | Nadghat           |
|                | 16           | RP2           | Raipur     | Arong             |
|                | 17           | DHM1          | Dhmtari    | Rampur            |
|                | 18           | MAH1          | Mahasamund | Saragoan          |
| Maharashtra    | 19           | KAW1          | Kawardha   | Praphayala        |
|                | 20           | DH1           | Dhule      | Arvi              |
|                | 21           | DH2           | Dhule      | Purmepanda        |
|                | 22           | AM2           | Amravati   | Dhonigoan         |
|                | 23           | AM3           | Amravati   | Chinchkheda       |
|                | 24           | AM4           | Amravati   | Anbhora           |

|         |    |       |            |             |
|---------|----|-------|------------|-------------|
|         | 25 | AK2   | Akola      | Hamgoan     |
|         | 26 | HIN2  | Hingoli    | Hingoli     |
|         | 27 | AUR5  | Aurangabad | Nandrabad   |
|         | 28 | AUR6  | Aurangabad | Hutnur      |
|         | 29 | PAR3  | Parbhani   | Jinturtabda |
|         | 30 | PAR4  | Parbhani   | Pedgoan     |
|         | 31 | PAR5  | Parbhani   | Bhogoan     |
|         | 32 | JAL 2 | Jalna      | Badnapur    |
|         | 33 | JAL 4 | Jalna      | Sindkhed    |
|         | 34 | JAL 5 | Jalna      | Dhanora     |
| Gujarat | 35 | SUR1  | Surat      | Songarh     |
|         | 36 | VAL1  | Valsad     | Pardhi      |
|         | 37 | VAD1  | Vadodara   | Ramegoan    |
|         | 38 | VAD2  | Vadodara   | Nanafaladia |
|         | 39 | DAH5  | Dahod      | Marijumgay  |
|         | 40 | DAH6  | Dahod      | Kalaharidi  |

Table. - 1 (B)

| S. No. | States          | Number of districts | Number of locations |
|--------|-----------------|---------------------|---------------------|
| 1      | Madhya Pradesh. | 6                   | 9                   |
| 2      | Chhattisgarh    | 6                   | 10                  |
| 3      | Maharashtra     | 7                   | 15                  |
| 4      | Gujarat         | 4                   | 6                   |
|        | <b>Total</b>    | <b>23</b>           | <b>40</b>           |

## **2. Isolation of *Rhizoctonia bataticola* from plant samples**

The roots of diseased chickpea plant showing root rot symptoms were washed thoroughly with tap water, small pieces from infected roots were cut with the help of sterilized blade. These pieces were surface sterilized with 1:1000 mercuric chloride ( $\text{HgCl}_2$ ) solution for one minute followed by three changes in sterilized distal water to remove trace of  $\text{HgCl}_2$ . The pieces were then transferred aseptically to petri plates containing PDA. Inoculated petri plates were incubated at  $28 \pm 1^\circ\text{C}$  for three to five days and examined at frequently intervals to see the growth of the fungus developing from different pieces.

## **3. Purification and identification of isolates of *Rhizoctonia bataticola***

The cultures of *R. bataticola* were purified by subculturing the single hyphal tip method and maintained by mass transfer on potato dextrose agar medium at room temperature. After purification these isolates of *R. bataticola* were identified by observing the colony against light with the naked eyes and later confirmed with the help of microscope.

## **4. Variability in isolates of *Rhizoctonia bataticola***

### **4.1 Cultural variability**

The cultural characters of isolates of *R. bataticola* were recorded from culture grown on PDA. Twenty ml of PDA was poured in each of previously sterilized petri plates. Five mm discs were cut through sterilized cork borer from the margin of seven days old colony of the fungal culture grown in petri plates. One disc was placed in the centre of each plate and incubated at  $28 \pm 1^\circ\text{C}$  for seven days. Three replications were maintained for each isolate.

## 4.2 Morphological variability

The slides of various isolates were prepared in lactophenol from 10 days old culture. For morphological studies 10 observations for size of sclerotia and size of hyphal width and number of sclerotia per microscopic field belonging to each isolates were taken under high power (40 x) microscopic field. The differences in colour of sclerotia and shape of sclerotia were observed. These isolates were categorized in various groups according to size of hyphal width, size of sclerotia, number of sclerotia, colour of sclerotia and shape of sclerotia.

## 4.3 Pathological variability

### 4.3.1 Pathogenicity test

Blotter paper technique as described by Nene *et al.* (1981): -

To conduct the pathogenicity test 20 surface sterilized (5 min in 2.5% sodium hypo chloride) seeds of the test line of susceptible variety BG 212 were sown in autoclaved riverbed sand in pots. The two-mycelial mats of *R. bataticola* were grown on P.D. Broth media for 5 days the then removed in 100 ml. sterilized distilled water, macerated in a warring blender for 1 min. The five days old seedlings were uprooted and washed the root system in running water followed by rinsing in sterilized distilled water. And then dipped in inoculum solution for 30 seconds up and down. After that inoculated seedlings were placed side by side on a blotter paper in two replications on tray and also maintain the line of control without culture. Placed the tray in an incubator at 35° C for 8 days and provided 12 hours artificial light. Moistened the blotters adequately every day or alternate day. At the end of incubation period (8 days), examined the seedlings for the extent of root damage and for the disease.

5. Screening of chickpea genotypes for host resistance under multiple disease sick field

In order to locate the sources of resistance, the chickpea lines were screened in multiple disease sick field which was prepared by adding the culture of *R. bataticola* in chickpea straw, sand medium.

The culture was mixed in the soil in the ratio of 1:9 and then added in each row before sowing. Fifty seeds of each entry were sown in 5-meter long row at 30 cm apart. A susceptible check JG 62 was maintained after every two rows of test entry for comparison.

Observations on per cent emergence was recorded after 20 days of sowing and mortality due to dry root rot were recorded weekly up to podding stage.

**Percent dry root rot:** Dry root rot percentage was calculated by the following formula.

Percent dry root rot = 
$$\frac{\text{Total number of DRR affected plants}}{\text{Total number of plants.}} \times 100$$

Table. -2: Disease rating scale

| Rating | Category             | DRR percentage (%) |
|--------|----------------------|--------------------|
| 1      | Resistant            | 0.0 - 10           |
| 2      | Moderately resistant | 10.1 - 20          |
| 3      | Tolerant             | 20.1 - 30          |
| 4      | Susceptible          | 30.1 - 40          |
| 5      | Highly susceptible   | And above 40       |

# RESULTS

## 1. Collection of diseased plant / root samples

Chickpea plants showing symptoms of dry root rot were procured from the Wilt Network Project (Chickpea) collected at flowering and pod formation stage from the 40 locations of 23 districts surveyed in four major chickpea growing states viz; Madhya Pradesh, Chhattishgarh, Maharashtra and Gujarat. The name of different localities along with other details are being presented as follows (Table. - 1)

## 2. Symptomatology

The first symptom of the disease is yellowing of the leaves within a day or two such leaves drop and in the course of the next two or three days whole plant dies off. The leaves and stem of the affected plants are usually showing straw coloured appearance. The taproot becomes dark and quite brittle in dry soil. The dark black sclerotia were observed and within host tissues when the stem of diseased plant is examined closely, dark lesion may be seen on the bark at the ground level. On uprooting of the plants from the soil, the basal stem and main root shows dry rot symptoms. The tissues are weakened and break off easily.

## 3. Isolation, purification and identification of *Rhizoctonia bataticola*

Isolation from infected chickpea plants showing symptoms of "Dry root rot" were made on potato dextrose agar medium. *Rhizoctonia bataticola* was found to be associated with plant / root of all the 40 collected samples from different localities.

The isolates were identified on the basis of cultural and morphological characters with the help of microscope. All the isolates of *Rhizoctonia bataticola* were purified by single hyphal tip technique and maintained by mass transfer on P.D.A. slants at 25 ° C for further investigations.

4. Cultural characters of different isolates of *Rhizoctonia bataticola*

4.1 Type of mycelial growth on P.D.A.

Cultural variability was studied on P.D.A medium and the characters observed on type of mycelial growth and colour of colony after 6 days of incubation at  $25 \pm 1^{\circ}\text{C}$  and growth rate of isolates of *Rhizoctonia bataticola* were recorded on alternate days. (Table. – 3, 4, 5 and Plate – 1)

On the basis of type of mycelial growth the isolates were categorized as fluffy, partial submerged and submerged. It is evident from the data presented in Table. - 3 that out of 40 isolates, 9 (22%) isolates were fluffy, 11 (28%) isolates were partial submerged and 20 (50%) isolates were having submerged type of mycelial growth. (Table. – 4)

Table. – 3: Type of mycelial growth and growth rate of various isolates of *Rhizoctonia bataticola* on potato dextrose agar medium at different intervals

| Isolates<br>No. | Type of mycelial<br>growth | Colour of<br>colony | Growth rate (mm.) * |                     |                     |
|-----------------|----------------------------|---------------------|---------------------|---------------------|---------------------|
|                 |                            |                     | 2 <sup>nd</sup> day | 4 <sup>th</sup> day | 6 <sup>th</sup> day |
| 1               | Submerged                  | Dark black          | 48                  | 90                  | -                   |
| 2               | Submerged                  | Grayish cottony     | 38                  | 77                  | 90                  |
| 3               | Submerged                  | Grayish cottony     | 59                  | 90                  | -                   |
| 4               | Partial submerged          | Grayish cottony     | 43                  | 71                  | 90                  |
| 5               | Partial submerged          | Dark black          | 54                  | 90                  | -                   |
| 6               | Fluffy                     | Dark black          | 55                  | 90                  | -                   |
| 7               | Submerged                  | Dark black          | 54                  | 90                  | -                   |
| 8               | Submerged                  | Dark black          | 38                  | 84                  | 90                  |
| 9               | Submerged                  | Dark black          | 48                  | 90                  | -                   |
| 10              | Submerged                  | Dark black          | 51                  | 90                  | -                   |

|    |                   |                 |    |    |    |
|----|-------------------|-----------------|----|----|----|
| 11 | Submerged         | Dark black      | 41 | 80 | 90 |
| 12 | Partial submerged | Grayish cottony | 42 | 87 | 90 |
| 13 | Fluffy            | Grayish cottony | 46 | 85 | 90 |
| 14 | Submerged         | Grayish cottony | 50 | 90 | -  |
| 15 | Fluffy            | Grayish cottony | 51 | 90 | -  |
| 16 | Partial submerged | Grayish cottony | 33 | 70 | 90 |
| 17 | Submerged         | Dark black      | 43 | 79 | 90 |
| 18 | Submerged         | Dark black      | 39 | 80 | 90 |
| 19 | Submerged         | Dark black      | 48 | 81 | 90 |
| 20 | Submerged         | Dark black      | 45 | 90 | -  |
| 21 | Partial submerged | Dark black      | 46 | 90 | -  |
| 22 | Submerged         | Dark black      | 47 | 80 | 90 |
| 23 | Partial submerged | Dark black      | 35 | 70 | 90 |
| 24 | Fluffy            | Dark black      | 40 | 90 | -  |
| 25 | Partial submerged | Dark black      | 49 | 88 | 90 |
| 26 | Submerged         | Dark black      | 29 | 71 | 90 |
| 27 | Submerged         | Dark black      | 39 | 79 | 90 |
| 28 | Submerged         | Dark black      | 47 | 87 | 90 |
| 29 | Partial submerged | Dark black      | 48 | 90 | -  |
| 30 | Fluffy            | Dark black      | 49 | 90 | -  |
| 31 | Fluffy            | Grayish cottony | 50 | 90 | -  |
| 32 | Fluffy            | Grayish cottony | 50 | 90 | -  |
| 33 | Partial submerged | Dark black      | 36 | 79 | 90 |
| 34 | Partial submerged | Grayish cottony | 50 | 90 | -  |
| 35 | Submerged         | Dark black      | 49 | 90 | -  |
| 36 | Fluffy            | Dark black      | 34 | 70 | 90 |
| 37 | Submerged         | Dark black      | 32 | 75 | 90 |
| 38 | Submerged         | Dark black      | 26 | 76 | 90 |
| 39 | Partial submerged | Grayish cottony | 37 | 78 | 90 |
| 40 | Fluffy            | Grayish cottony | 48 | 90 | -  |

\* Average of three replications

Table. – 4: Grouping of isolates of *Rhizoctonia bataticola* on the basis of type of mycelial growth

| Mycelial growth   | Isolates                                              | Total |
|-------------------|-------------------------------------------------------|-------|
| Fluffy            | 6,13,15,24,30,31,32,36,40                             | 9     |
| Partial submerged | 4,5,12,16,21,23,25,29,33,34,39                        | 11    |
| Submerged         | 1,2,3,7,8,9,10,11,14,17,18,19,20,22,26,27,28,35,37,38 | 20    |
| Total             |                                                       | 40    |

4.2 Colour of colony

colour of colony was also observed for each isolates and categorized into two groups as dark black colour and grayish cottony. Out of the 40 isolates 27 (67.5%) Isolates were found as dark black colour and 13 (32.5%) isolates were found as grayish cottony colour of colony. (Table. - 12)

Table. – 5: Grouping of isolates of *Rhizoctonia bataticola* on the basis of colour of colony

| Characters      | Isolates                                                                 | Total |
|-----------------|--------------------------------------------------------------------------|-------|
| Dark black      | 1,5,6,7,8,9,10,11,17,18,19,20,21,22,23,25,26,27,28,29,30,33,35,36,37,38, | 27    |
| Grayish cottony | 2,3,4,12,1,3,14,15,16,31,32,34,39,40,                                    | 13    |
|                 | Total                                                                    | 40    |

4.3 Growth rate on P.D.A.

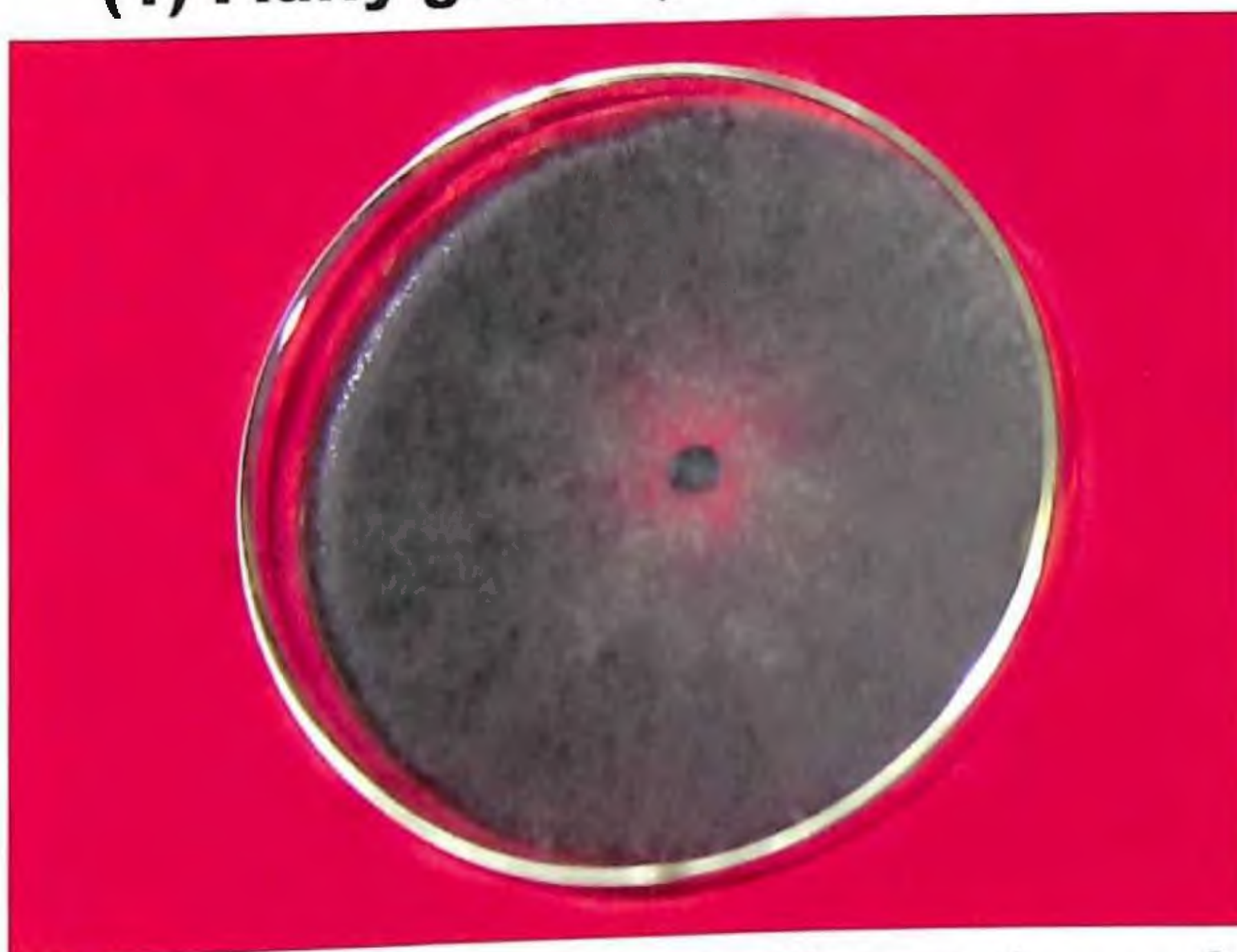
Studies on growth rate of isolates was conducted to know the fast growing, medium growing and slow growing isolates at different intervals on P.D.A. Observations on growth rate of different isolates were made after 2<sup>nd</sup>, 4<sup>th</sup> and 6<sup>th</sup> day of incubation. On the basis of 4<sup>th</sup> day observation the data denotes that the slow growth (70-76 mm) has been obtained in isolates 4,16,23,26,36,37,38 and isolates 2,11,17,18,19,22,27,33,39 showed medium growth (77-80 mm). Rest of the isolates attained the fast growth (84-90 mm). The isolate no. 3 belong to Dindori had maximum growth 59 mm. while the minimum growth 26 mm. was observed in isolate no. 38 which is from Vadodara (Table. – 5).

Table. - 6: Grouping of isolates of *Rhizoctonia bataticola* on the basis of growth rate.

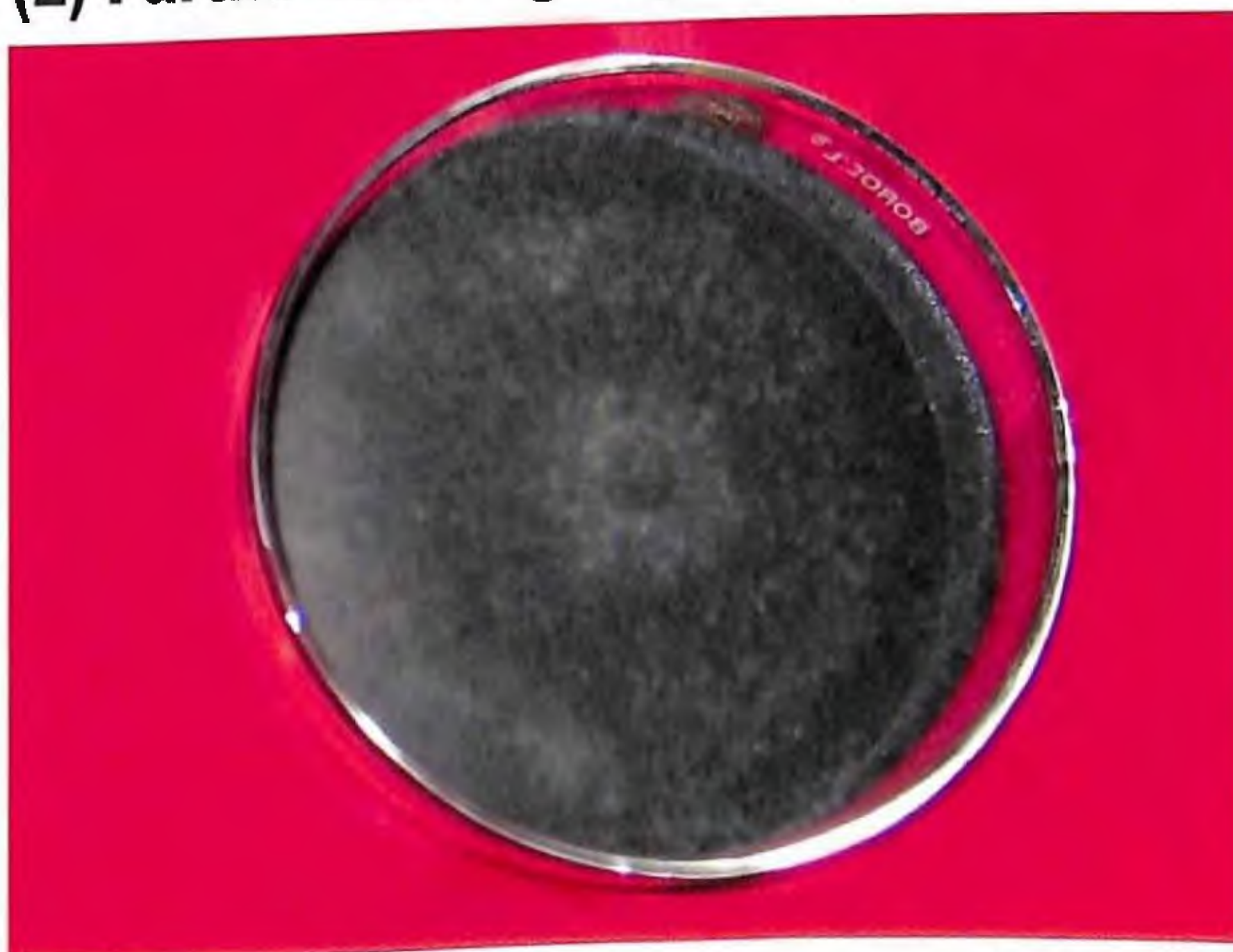
| Characters               | Isolates                                                             | Total |
|--------------------------|----------------------------------------------------------------------|-------|
| Slow growth (70-76 mm)   | 4,16,23,26,36,37,38                                                  | 7     |
| Medium growth (77-83 mm) | 2,11,17,18,19,22,27,33,39,                                           | 9     |
| Fast growth (84-90 mm)   | 1,3,5,6,7,8,9,10,12,13,14,15,20,21,24,25,<br>28,29,30,31,32,34,35,40 | 24    |
|                          | Total                                                                | 40    |



**(1) Fluffy growth, isolate no - 31**



**(2) Partial submerged growth, isolate no - 2**



**(3) Submerged growth, isolate no - 7**

**Plate-1: Type of mycelial growth of various isolates of *Rhizoctonia bataticola* on**

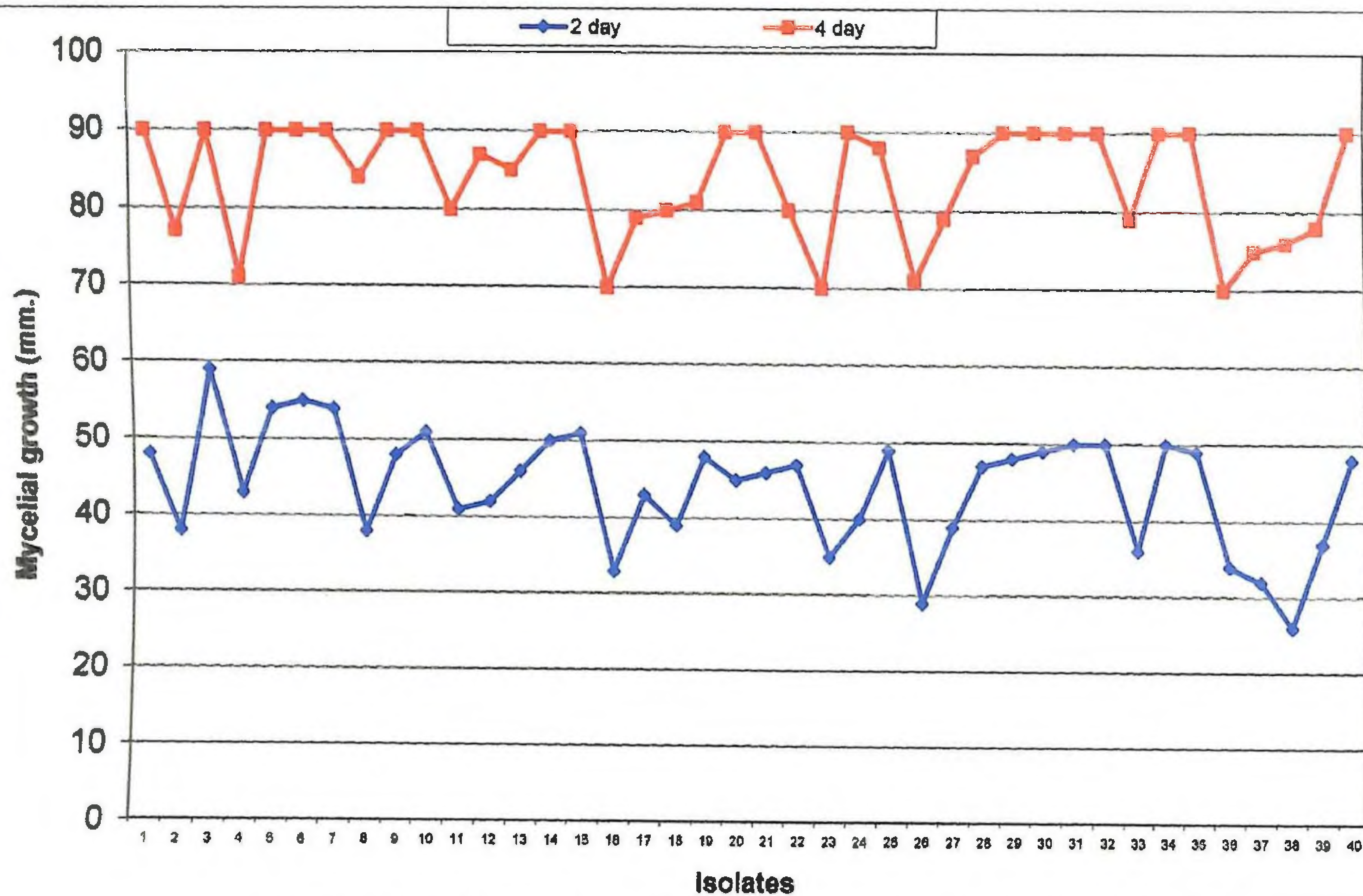


Figure - 1: Growth rate of isolates of *Rhizoctonia bataticola* at different intervals

## 5. Morphological characters of different isolates of *Rhizoctonia bataticola*

To know the variability in morphological characters of different isolates of *Rhizoctonia bataticola* viz, size of hyphal width, size of sclerotia, number of sclerotia colour of sclerotia and shape of sclerotia have been studied and summarized in Table. – 6. Ten Microscopic observations under high power (40 X) were carried out for each isolate and the average value is presented. (Table. – 6, 7, 8, 9, 10, 11 and Plate – 2)

Table. - 7: Morphological characters of various isolate of *Rhizoctonia bataticola*

| Isolate No | Size of hyphal width (µm) | Size of Sclerotia (µm) | Number of Sclerotia / Microscopic field | Colour of Sclerotia | Shape of Sclerotia |
|------------|---------------------------|------------------------|-----------------------------------------|---------------------|--------------------|
| 1          | 5.6                       | 89.0 X 85.2            | 11.7                                    | Dark brown          | Round              |
| 2          | 7.6                       | 103.3 X 90.2           | 08.1                                    | Dark brown          | Oblong             |
| 3          | 6.7                       | 98.2 X 88.2            | 12.6                                    | Brown               | Round              |
| 4          | 5.7                       | 79.3 X 69.7            | 16.2                                    | Brown               | Oblong             |
| 5          | 7.6                       | 73.0 X 62.1            | 18.8                                    | Dark brown          | Round              |
| 6          | 7.4                       | 77.6 X 70.7            | 22.3                                    | Dark brown          | Oblong             |
| 7          | 6.5                       | 87.5 X 68.2            | 14.0                                    | Brown               | Oblong             |
| 8          | 7.2                       | 99.1 X 76.9            | 17.2                                    | Dark brown          | Round              |
| 9          | 5.5                       | 102.2 X 73.6           | 09.2                                    | Black               | Round              |
| 10         | 7.6                       | 84.2 X 72.4            | 12.8                                    | Dark brown          | Round              |
| 11         | 7.0                       | 92.8 X 86.8            | 10.8                                    | Black               | Round              |
| 12         | 9.3                       | 92.1 X 85.5            | 18.7                                    | Black               | Oblong             |
| 13         | 5.2                       | 74.5 X 64.0            | 19.2                                    | Brown               | Oblong             |

|    |     |               |      |            |        |
|----|-----|---------------|------|------------|--------|
| 14 | 6.8 | 83.6 X 71.2   | 14.8 | Dark brown | Round  |
| 15 | 7.6 | 87.8 X 77.1   | 16.1 | Dark brown | Round  |
| 16 | 8.2 | 78.5 X 67.2   | 19.2 | Dark brown | Round  |
| 17 | 7.5 | 73.8 X 63.9   | 12.8 | Dark brown | Round  |
| 18 | 6.2 | 91.5 X 85.3   | 12.7 | Dark brown | Round  |
| 19 | 6.6 | 103.7 X 90.1  | 09.7 | Black      | Round  |
| 20 | 6.4 | 117.2 X 104.0 | 10.2 | Dark brown | Round  |
| 21 | 6.1 | 113.2 X 106.5 | 08.1 | Dark brown | Round  |
| 22 | 5.7 | 88.7 X 79.5   | 11.0 | Dark brown | Round  |
| 23 | 5.7 | 72.7 X 59.6   | 17.9 | Dark brown | Round  |
| 24 | 7.2 | 106.2 X 90.2  | 11.4 | Brown      | Round  |
| 25 | 7.6 | 105.7 X 99.2  | 08.5 | Dark brown | Oblong |
| 26 | 7.8 | 82.3 X 72.3   | 10.8 | Dark brown | Round  |
| 27 | 6.4 | 87.5 X 71.7   | 14.4 | Black      | Round  |
| 28 | 5.7 | 87.8 X 76.3   | 13.7 | Black      | Round  |
| 29 | 6.5 | 89.9 X 75.7   | 12.6 | Dark brown | Oblong |
| 30 | 5.2 | 78.8 X 75.1   | 16.5 | Dark brown | Round  |
| 31 | 5.4 | 76.2 X 65.9   | 18.5 | Dark brown | Oblong |
| 32 | 6.8 | 89.9 X 79.4   | 12.1 | Black      | Round  |
| 33 | 7.0 | 90.8 X 73.8   | 10.6 | Black      | Round  |
| 34 | 6.2 | 89.2 X 76.1   | 14.2 | Black      | Oblong |
| 35 | 8.7 | 92.8 X 76.0   | 13.5 | Dark brown | Round  |
| 36 | 7.1 | 100.4 X 76.2  | 10.2 | Dark brown | Round  |
| 37 | 5.6 | 99.0 X 89.3   | 09.9 | Dark brown | Round  |
| 38 | 7.1 | 86.7 X 89.0   | 11.1 | Dark brown | Round  |
| 39 | 5.8 | 85.6 X 73.6   | 10.0 | Brown      | Oblong |
| 40 | 6.6 | 96.0 X 77.1   | 11.3 | Black      | Oblong |

\* Average of 10 observations.

5.1 Size of hyphal width

It is evident from the data presented in Table. - 7 that wide range of variations has been observed in size of hyphal width.

Average of 10 microscopic observations on the size of hyphal width varied from 5.2-9.3  $\mu\text{m}$ . On the basis of observations on size of hyphal width the isolates were classified into three categories viz., very thin (5.2-6.5  $\mu\text{m}$ ), thin (6.6-7.9  $\mu\text{m}$ ) and thick (8.0-9.3  $\mu\text{m}$ ). Out of 40 isolates, 18 (45%) isolates come under very thin size of hyphal width, 19 (48.5%) isolates comes under thin size of hyphal width, and 3 (7.5%) isolates comes under thick size of hyphal width. Among all the isolates, the isolate no. 12 belong to Bilaspur had maximum size of hyphal width 9.3  $\mu\text{m}$ . while the minimum size of hyphal width 5.2  $\mu\text{m}$ . was observed in isolate no. 30 which is belong to Parbhani. The *Rhizoctonia* isolates consist of barrel shaped cells, mycelium branched and light to dark brown in colour, septation within the cell also varied from one isolate to another isolate. (Table. – 7)

Table. - 8: Grouping of isolates of *Rhizoctonia bataticola* on the basis of size of hyphal width

| Characters                         | Isolates                                               | Total |
|------------------------------------|--------------------------------------------------------|-------|
| Very thin (5.2-6.5 $\mu\text{m}$ ) | 1,4,7,9,13,18,20,21,22,23,27,28,29,30,31,34,37,39      | 18    |
| Thin (6.6-7.9 $\mu\text{m}$ )      | 3,5,6,8,10,11,14,15,17,19,24,25,26, 32,33,34,36,38, 40 | 19    |
| Thick (8.0-9.3 $\mu\text{m}$ )     | 12,16,35                                               | 3     |
|                                    | Total                                                  | 40    |

5.2 Size of sclerotia

The data presented in Table 6 denote that on the basis of 10 Microscopic observations the size of sclerotia varied from 72.7-117.2 x 57.1-106.5 µm. The isolates were classified into three groups, viz; minute size (72.7-87.5 x 57.1-73) medium size (87.6-103.2 x 73.5-90.0) and bold size (103.3-117.2 x 90.1-106.5). Out of 40 isolates, 15 (37.5%) of isolates were minute, 19 (47.5%) isolates were medium size and 6 (15%) were of bold size sclerotia. Among all the isolates, the isolate no. 20 belong to Dhule had maximum bold size of sclerotia 117.2 X 104.0 µm. while the very Minute size of sclerotia 72.7 X 59.6 µm. was observed in isolate no. 23 which is belong to Amravati (Table. - 8)

Table. – 9: Grouping of isolates of *Rhizoctonia bataticola* on the basis of size of sclerotia.

| Characters                            | Isolates                                                   | Total |
|---------------------------------------|------------------------------------------------------------|-------|
| Minute<br>(72.7-87.5 X 57.1-73.5 µm)  | 4,5,6,7,10,13,14,16,17,23,26,27,30,31,39                   | 15    |
| Medium<br>(87.6-103.2 X 73.5-90.0 µm) | 1,3,8,9,11,12,15,18,22,28,29,32,33,34,35,<br>36, 37,38,40, | 19    |
| Bold<br>(103.3-117.2 X90.1-106.5 µm)  | 2,19,20,21,24,25                                           | 6     |
|                                       | Total                                                      | 40    |

5.3 Number of sclerotia

Number of sclerotia per microscopic field was also observed for each isolate and grouped into, three groups viz; minimum (8 -12), medium (13-17), maximum (18-22). The number of sclerotia varied from 8.1 to 22.3. Out of 40 isolates 23 (57.5%) isolates were observed under minimum number of sclerotia, 11 (27.5%) isolates were encountered under medium number of sclerotia and 6 (15%) isolates were categorized under maximum number of sclerotia. Among all the isolates, the isolate no. 6 belong to Betul had maximum number of sclerotia

22.3 while the minimum number of sclerotia 8.1 was observed in isolate no. 21 which is belong to Dhule (Table. – 9)

Table. – 10: Grouping of isolates of *Rhizoctonia bataticola* on the basis of number of sclerotia

| Characters      | Isolates                                                         | Total |
|-----------------|------------------------------------------------------------------|-------|
| Minimum (8-13)  | 1,2,3,9,10,11,17,18,19,20,21,22,24,25,26,29,32,33,36,37,38,39,40 | 23    |
| Medium (13-18)  | 4,7,8,14,15,23,27,28,30,34,35                                    | 11    |
| Maximum (18-23) | 5,6,12,13,16,31                                                  | 6     |
|                 | Total                                                            | 40    |

#### 5.4 Colour of sclerotia

Colour of sclerotia was also observed and categorized into three groups as brown, dark brown and black. Out of the 40 isolates 6 (15%) Isolates were found as brown, 24 (60%) isolates were found as dark brown and 10 (25%) isolates were found as black colour of sclerotia (Table. - 10)

Table. – 11: Grouping of isolates of *Rhizoctonia bataticola* on the basis of colour of sclerotia

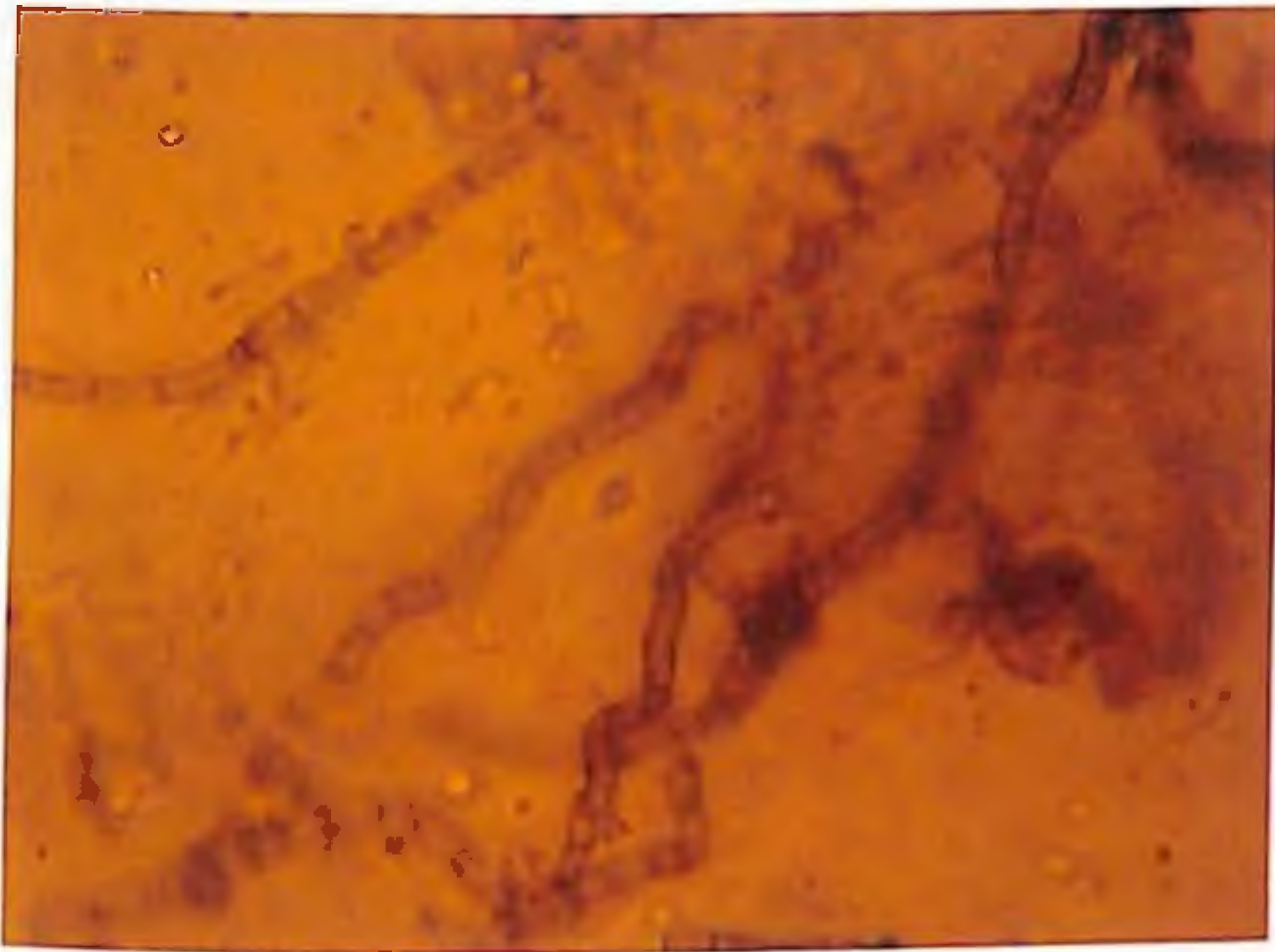
| Characters | Isolates                                                           | Total |
|------------|--------------------------------------------------------------------|-------|
| Brown      | 3,4,7,13,24,39                                                     | 6     |
| Dark brown | 1,2,5,6,8,10,14,15,16,17,18,20,21,22,23,25,26,29,30,31,35,36,37,38 | 24    |
| Black      | 9,11,12,19,27,28,32,33,34,40                                       | 10    |
|            | Total                                                              | 40    |

5.5 Shape of sclerotia

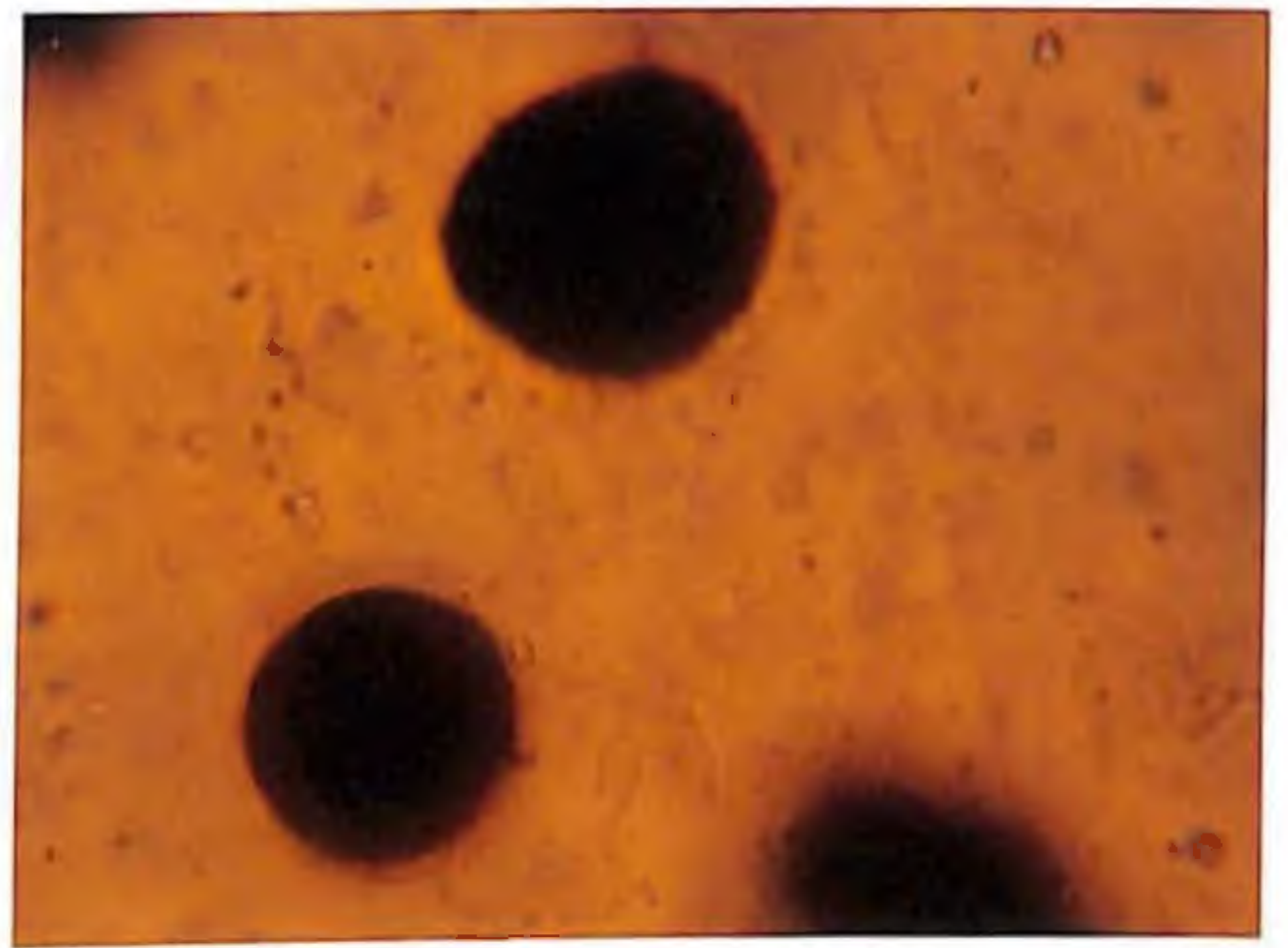
Shape of sclerotia was also observed for each isolates and categorized into two groups as oblong shape and round shape. Out of the 40 isolates 12 (30%) Isolates were found as oblong shape and 28 (70%) isolates were found as round shape of sclerotia. (Table. - 11)

Table. – 12: Grouping of isolates of *Rhizoctonia bataticola* on the basis of shape of sclerotia

| Characters   | Isolates                                                                          | Total |
|--------------|-----------------------------------------------------------------------------------|-------|
| Oblong shape | 2,4,6,7,12,13,25,29,31,34,39,40                                                   | 12    |
| Round shape  | 1,3,5,8,9,10,11,12,14,15,16,17,18,19,20,21,22,23,24,26,27,28,30,32,33,35,36,37,38 | 28    |
|              | Total                                                                             | 40    |



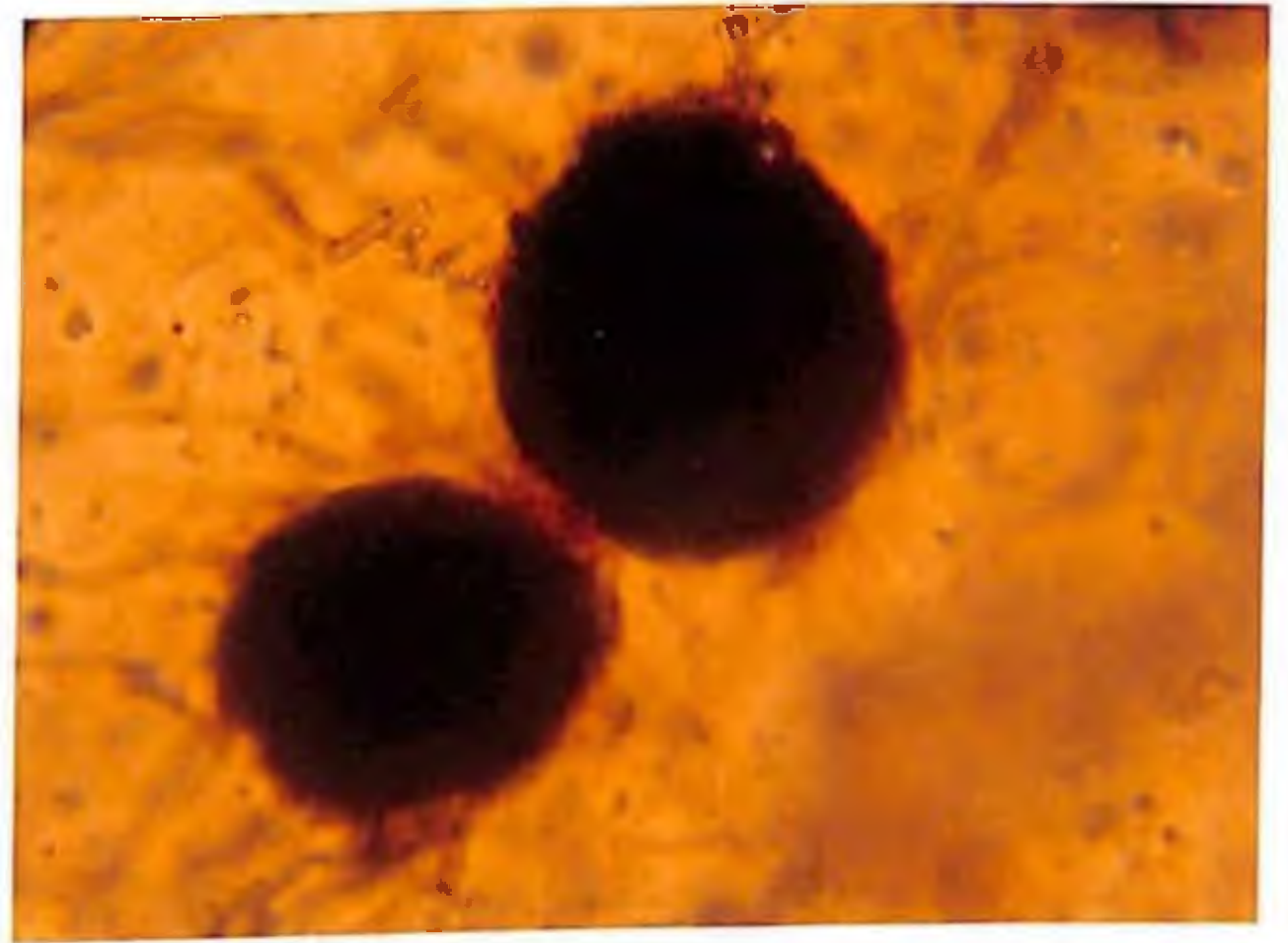
**(1) Very thin, isolate no - 30**



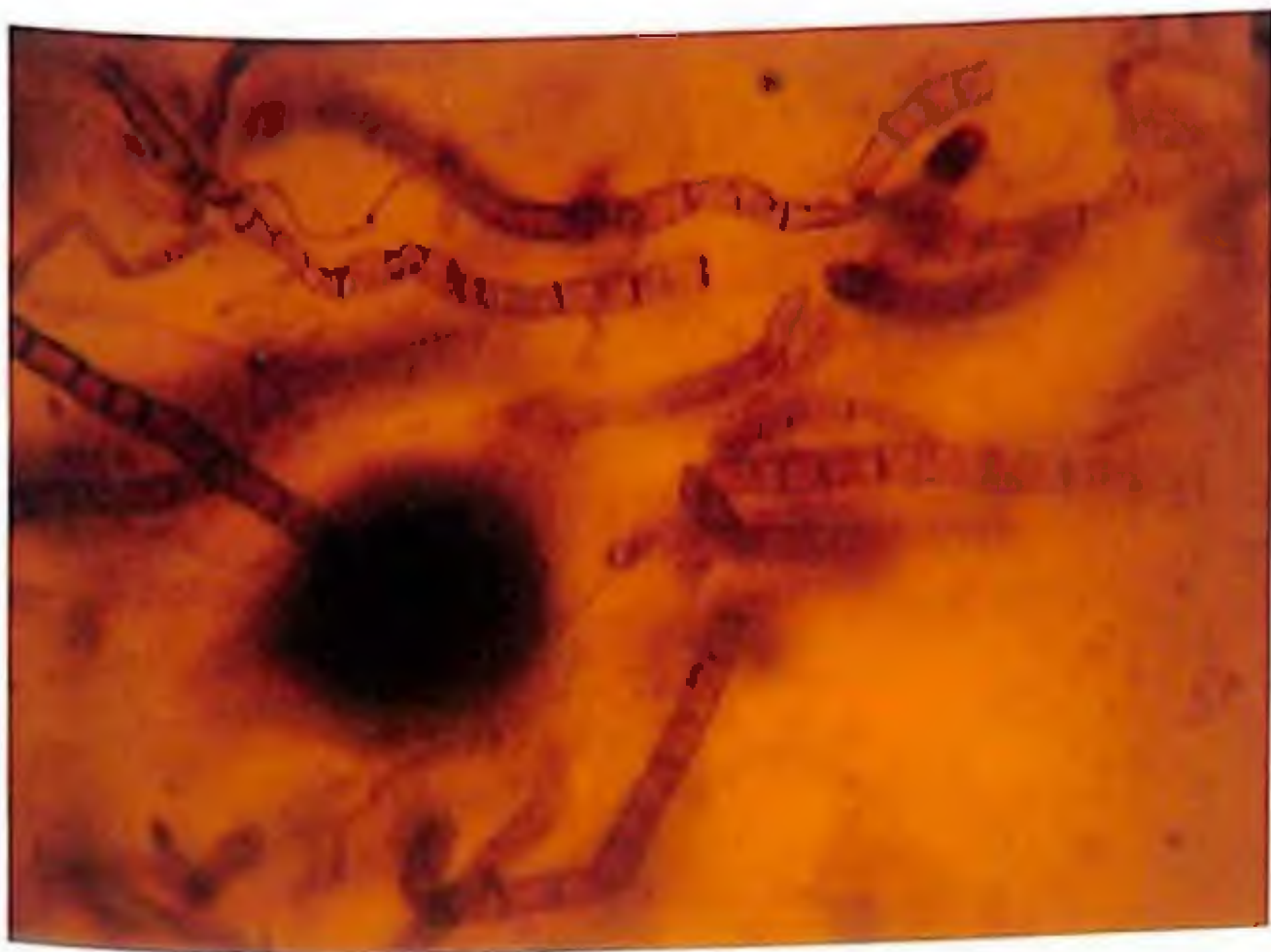
**(1) Minute size, isolate no - 23**



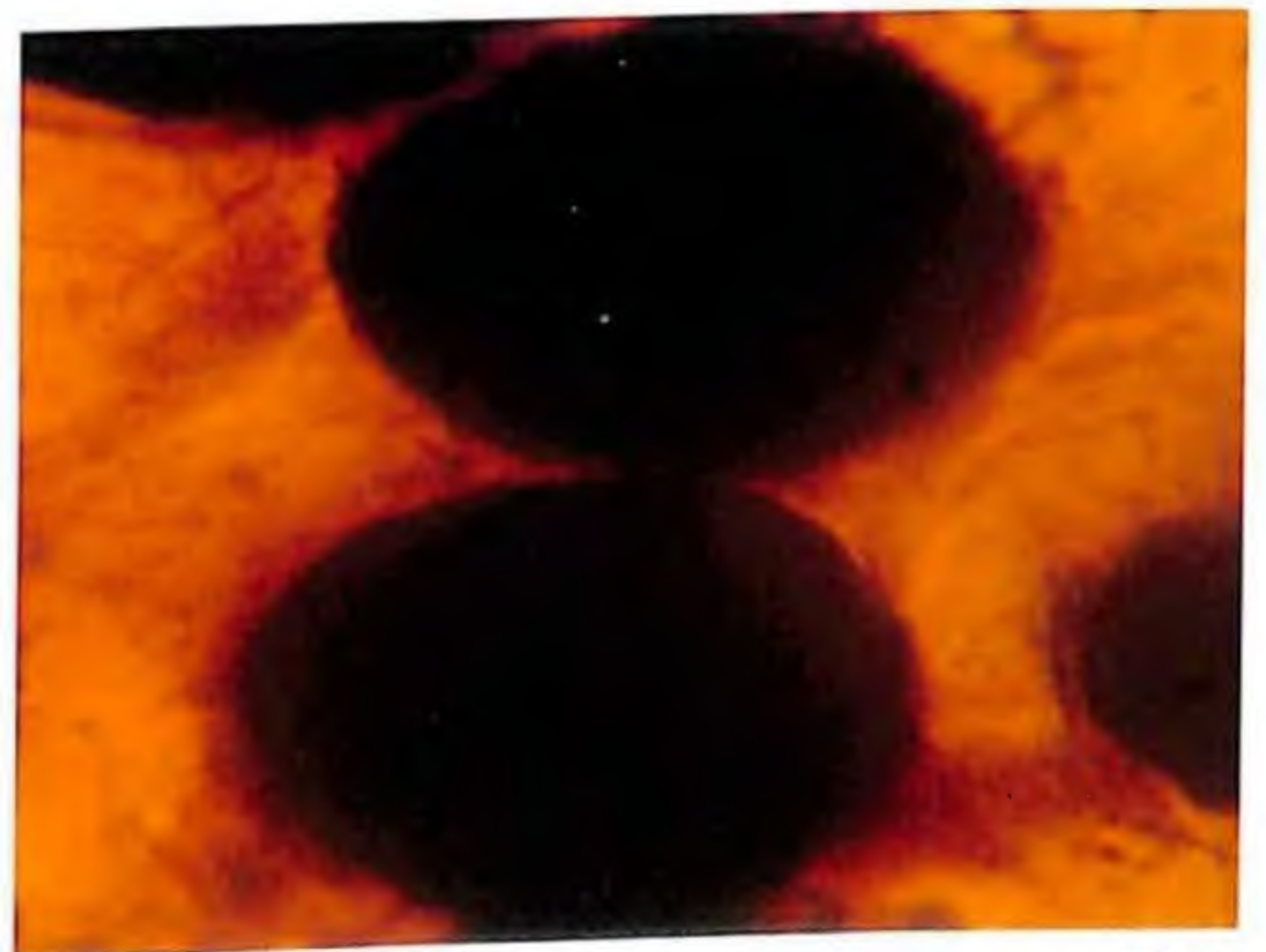
**(2) Thin size, isolate no - 18**



**(2) Medium size, isolate no - 8**



**(3) Thick size, isolate no - 12**



**(3) Bold size, isolate no - 20**

**Variation in size of hyphal width**

**Variation in size of sclerotia**

6. Pathological variability

Blotter paper technique as described by Nene *et al.* (1981) was employed to know the pathological variability among the isolates of *Rhizoctonia bataticola*. A highly susceptible chickpea variety BG-212 was used to see the disease rating and maximum mortality causing after inoculation by the isolates. (Table. - 12)

Table. - 13: Pathological characters of various isolate of *Rhizoctonia bataticola*

| Isolates no. | Disease rating | Category        |
|--------------|----------------|-----------------|
| 1            | 8.3            | Highly virulent |
| 2            | 8.4            | Highly virulent |
| 3            | 8.7            | Highly virulent |
| 4            | 7.9            | Virulent        |
| 5            | 8.4            | Highly virulent |
| 6            | 8.7            | Highly virulent |
| 7            | 7.3            | Virulent        |
| 8            | 8.6            | Highly virulent |
| 9            | 8.2            | Highly virulent |
| 10           | 6.6            | Virulent        |
| 11           | 6.7            | Virulent        |
| 12           | 8.0            | Highly virulent |
| 13           | 8.8            | Highly virulent |
| 14           | 8.7            | Highly virulent |
| 15           | 7.8            | Virulent        |
| 16           | 8.8            | Highly virulent |
| 17           | 7.2            | Virulent        |
| 18           | 8.5            | Highly virulent |
| 19           | 7.5            | Virulent        |

|    |     |                 |
|----|-----|-----------------|
| 20 | 8.7 | Highly virulent |
| 21 | 8.4 | Highly virulent |
| 22 | 8.4 | Highly virulent |
| 23 | 8.7 | Highly virulent |
| 24 | 8.3 | Highly virulent |
| 25 | 9.0 | Highly virulent |
| 26 | 7.3 | Virulent        |
| 27 | 8.2 | Highly virulent |
| 28 | 8.1 | Highly virulent |
| 29 | 8.1 | Highly virulent |
| 30 | 8.4 | Highly virulent |
| 31 | 8.9 | Highly virulent |
| 32 | 8.6 | Highly virulent |
| 33 | 8.5 | Highly virulent |
| 34 | 8.6 | Highly virulent |
| 35 | 8.5 | Highly virulent |
| 36 | 7.4 | Virulent        |
| 37 | 8.3 | Highly virulent |
| 38 | 8.6 | Highly virulent |
| 39 | 8.3 | Highly virulent |
| 40 | 7.3 | Virulent        |

Observation on disease rating was recorded after 8<sup>th</sup> day of incubation on 20 seedlings. The isolates varied in causing the infection on variety BG 212, as it is evident from the data presented in table 12. On the basis of disease score the isolates were categorized into two groups viz., virulent and highly virulent. The disease rating of isolates were varied from 6.6-9.0. Out of 40 isolates 10 (25%) isolates were found virulent, as the disease rating score was 6.6-8.0 whereas 30 (75%) isolates categorized as highly virulent. Where the disease rating was observed ranging from 8.1-9.0. Two pathotypes were identified on the basis of

pathogenicity test. Among all the isolates, the isolate no. 25 belong to Akola had maximum Disease rating 9.0 while the minimum Disease rating 6.6 was observed in isolate no. 10 which is belong to Raigarh (Table. – 9)

Table. – 14: Grouping of isolates of *Rhizoctonia bataticola* on the basis of pathogenicity

| Category        | Isolates                                                                           | Total |
|-----------------|------------------------------------------------------------------------------------|-------|
| Virulent        | 4,7,10,11,15,17,19,26,36,40                                                        | 10    |
| Highly virulent | 1,2,3,5,6,8,9,12,13,14,16,18,20,21,22,23,24,25,27,28,29,30,31,32,33,34,35,37,38,39 | 30    |
| Total           |                                                                                    | 40    |



**(1) Disease rating - 5, isolate no. - 7**



**(2) Disease rating - 7, isolate no. - 15**

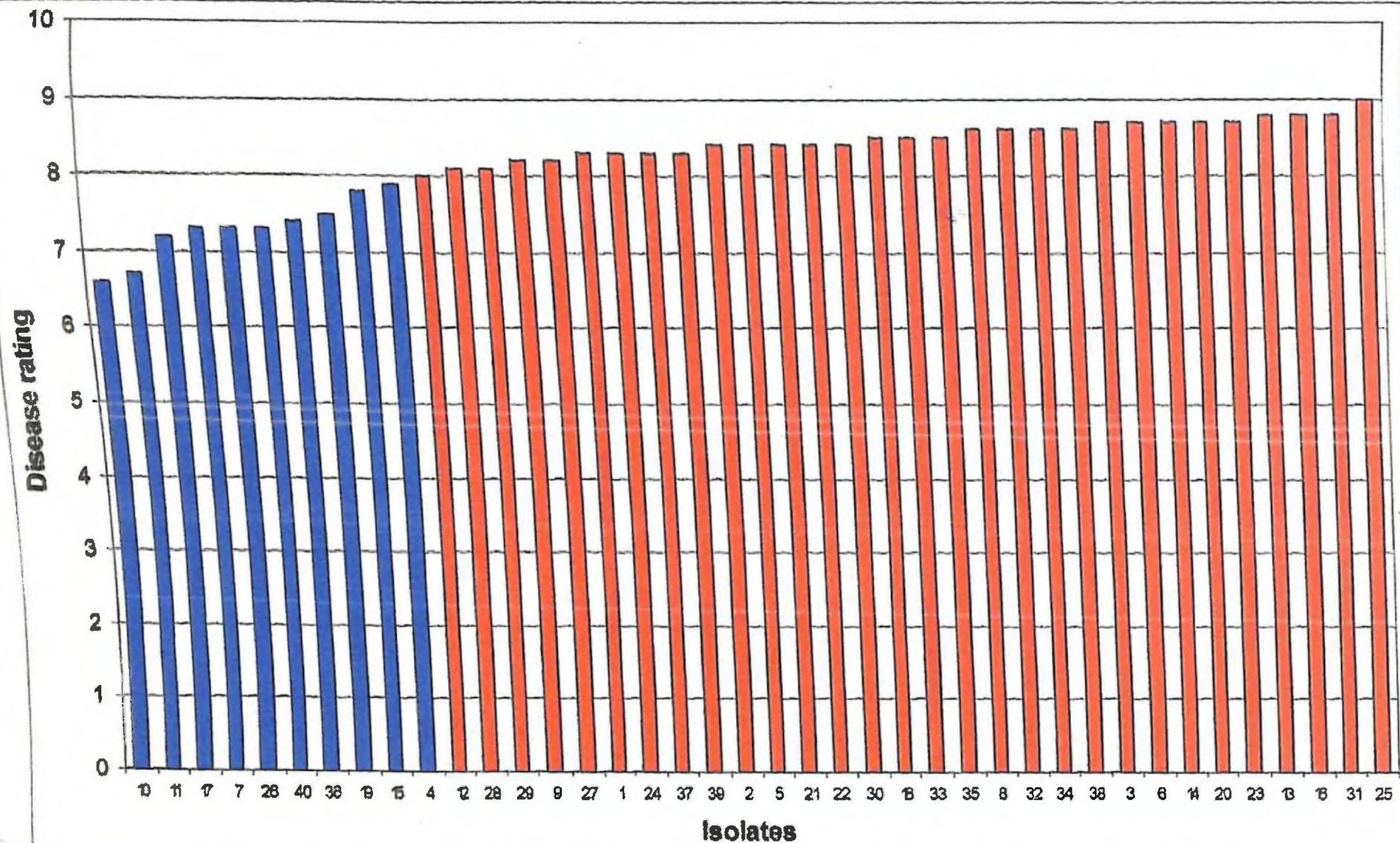


**(3) Disease rating - 9, isolate no. - 25**



**Un inoculated**

**Plate - 3: Chickpea variety BG - 212 sowing variability in disease rating, inoculated by different isolates of *Rhizoctonia bataticola*.**



Figur - 2: Reaction of different isolates of *Rhizoctonia bataticola* on chickpea variety BG 212



|     |            |       |
|-----|------------|-------|
| 4.  | BCP - 91   | 10.55 |
| 5.  | BG - 2041  | 23.20 |
| 6.  | BGD - 72   | 20.1  |
| 7.  | BGD - 130  | 9.00  |
| 8.  | BGD - 133  | 33.40 |
| 9.  | BGD - 1031 | 15.57 |
| 10. | BGD - 9812 | 8.39  |
| 11. | CSJ - 140  | 6.06  |
| 12. | CSJ - 313  | 6.45  |
| 13. | CSJ - 346  | 24.7  |
| 14. | CSJ - 313  | 7.50  |
| 15. | CSJ - 417  | 26.5  |
| 16. | CSJ - 426  | 19.2  |
| 17. | CSJ - 427  | 28.8  |
| 18. | CSJ - 428  | 13.00 |
| 19. | CSJ - 429  | 19.40 |
| 20. | CSJ - 431  | 25.70 |
| 21. | CSJ - 437  | 8.20  |
| 22. | CSJ - 438  | 53.80 |
| 23. | CSJ - 440  | 15.00 |
| 24. | CSJ - 441  | 19.40 |
| 25. | CSJ - 449  | 2.00  |
| 26. | CSJ - 453  | 11.50 |
| 27. | CSJ - 456  | 29.10 |
| 28. | CSJ - 458  | 18.90 |
| 29. | CSJ - 463  | 10.20 |
| 30. | CSJ - 464  | 20.00 |
| 31. | CSJ - 469  | 12.80 |
| 32. | CSJ - 470  | 17.70 |
| 33. | CSJ - 471  | 25.00 |
| 34. | CSJK - 3   | 20.69 |
| 35. | GL - 22044 | 8.5   |

Table. –16: The entries / varieties can be graded on the basis of percent mortality due to dry root rot.

| Dry root rot<br>Percent (%) | Entries                                                                                                                                                                                                                                                                                                                                                                               | Grading                 | No. of<br>entries |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-------------------|
| 0.0 to 10                   | BCP-21, BGD-130, BGD-9812, CSJ-140, CSJ-313, CSJ-437, CSJ-449, GL-22044, IPC-2000-25, MPJG-2000-25, MPJG-2001-99-115.                                                                                                                                                                                                                                                                 | Resistant               | 11                |
| 10.1 to 20                  | BCP-8, BCP - 91, BGD-1031, CSJ-426, CSJ-428, CSJ-429, CSJ-440, CSJ-440, CSJ - 441, CSJ-453, CSJ-458, CSJ-463, CSJ-464, CSJ-469, CSJ-469, CSJ-470, GNG-1602, GNG-1614, GNG-1751, H-02-36, H-02-125, IPC-2004-52, IPC-2005-56, IPC-2005-65, IPC-97-22, IPC-2K-3-27, JG-2003-14-16, JG-2003-23, JG-2003-01101, JSC-38, K-850, MPJG-2001-4, MPJG-2002-108, PG-97-08-6, PUSA-256, RSG-931. | Moderately<br>Resistant | 36                |
| 20.1 to 30                  | BCP-49, BG-2041, CSJ-346, CSJ-417, CSJ-427, CSJ-431, CSJ-456, CSJ-456, CSJ-471, CSJK-3, GNG-1618, IPC-2002-120, MPJG-89-9023.                                                                                                                                                                                                                                                         | Tolerant                | 13                |
| 30.1 to 40                  | BGD-133, PUSA-362, TCS-27.                                                                                                                                                                                                                                                                                                                                                            | Susceptible             | 03                |
| Above to 40                 | CSJ - 438, JG - 74.                                                                                                                                                                                                                                                                                                                                                                   | Highly<br>Susceptible   | 02                |
|                             | <b>Total</b>                                                                                                                                                                                                                                                                                                                                                                          |                         | <b>65</b>         |

## DISCUSSION

In Indian agriculture pulses play an important role not only from economical point of view but also due to their nutritional value. The crop suffers due to many diseases of which dry root rot caused by *Rhizoctonia bataticola* is the most important as heavy losses are caused by this disease. The disease causes plant mortality in podding and maturity stage of the crop growth when the temperature exceeds  $>25-30^{\circ}\text{C}$ .

Present investigations were undertaken with a view to know the variability among isolates of *R. bataticola* (cultural, morphological, and pathological variability) collected / procured from 40 locations of different districts of Madhya Pradesh, Chhattisgarh, Maharashtra, and Gujarat and to correlate them with virulence of isolates. A very little information on variability among isolates of *R. bataticola* is available in the literature particularly in chickpea dry root rot.

During collection of the dry root rot infected plants, the disease was found to occur flowering and podding stage of the crop in the range of 3.5 to 20.6 per cent. The dry root rot symptoms observed in diseased plant exhibited straw colour appearance and taproot mainly becomes brittle having minute sclerotia. These symptoms separated this disease from the so-called "Wilt complex" as stated by various workers in India. (Nene, 1989). Maximum atmospheric temperature up to  $34.5^{\circ}\text{C}$  favoured the disease development. Substantial mortality and loss in chickpea due to *R. bataticola* had been reported in high ambient temperature  $30^{\circ}\text{C}$  and above in the post – flowering stage (Singh and Mehrotra 1982).

Forty isolates as pure culture of *R. bataticola* were isolated from infected plants / root sample from different districts of Madhya Pradesh, Chhattisgarh, Maharashtra and Gujarat and were designated, as isolates number 1 to 40.

Isolates of *R. bataticola* were grouped on the basis of cultural, morphological and pathological characters.

Cultural characters of all the isolates of *R. bataticola* on P.D.A. medium showed variation within the isolates. Grouping of isolates on the basis of type of growth viz., fluffy, submerged and partial submerged. Indicates the possibility of three groups within the isolates. These results confirm the observations made by Byadgi and Hegde (1985) who reported fluffy, dark brown mycelium, partial fluffy mycelium and appressed light brown mycelium growth on different isolates of *R. bataticola*. Other workers have also reported the variation in type of growth isolates of *R. bataticola* (Sobti and Bansal, 1988; Ratnoo *et al.* 1997; Hooda and Grover, 1982 ;)

colour of colony was also observed for each isolates and categorized into two groups as dark black and grayish cottony. Out of the 40 isolates 27 (67.5%) Isolates were found as dark black colour and 13 (32.5%) isolates were found as grayish cottony colour of colony. The above findings was also reported by Shekhar *et al.* (2006) they found that dark black, grayish white, cottony white and grayish black colour of colony in his investigation.

After critically going through the data on growth rate recorded after 24<sup>hr</sup> intervals concluded that the isolates were fast growing (24 isolates), medium growing (9 isolates) and slow growing (7 isolates), which indicate differential growth rate in isolates. The observations coincide with Jharia (1985) who observed variation in growth rate of *R. bataticola*. The soil isolate was fast (90mm) followed by root (89mm) and seed (67mm) isolates on potato dextrose agar medium. Monga and Sheo Raj (1994) who observed that Rs-2 group of isolates was fastest in mycelial growth and Rs-4 was slow growing among all the four groups of isolates on potato dextrose agar medium. Byadgi and Hegde (1985) also reported fast growth and slow growth in his investigation.

Morphological characters of all the isolates showed variations within isolates. The size of hyphal width grouped as very thin, thin and thick which ranged from 5.2-9.3 $\mu$ m. thin type of hyphal width being the highest as the maximum number of isolates (48.5%) belongs to this category. Present observations coincide with the observations of Sobti and Sharma (1992) who reported that the width of hyphae varied from 4.16-8.48 $\mu$ m. Jharia (1985) observed the hyphal width range between 3.00-4.50 $\mu$ m.

The size of sclerotia of the isolates also has a wide range of variation from 72.7-117.2 x 57.1-106.5 $\mu$ m and thus categorized as minute, medium and bold type. Medium type being the highest, as the maximum number of isolates 47.5% belongs to this category. The above findings were also reported that the sclerotial size range between 66.14-128.25 $\mu$ m Mandal *et al.* (1998), 122-188 x 77-127 $\mu$ m from root isolates. Dhingra and Sinclair (1973), 84-93  $\mu$ m Jharia (1985), 73.88-188.33 x 51.09-155.90  $\mu$ m Sobti and Sharma (1992), 60-165 X 57-114  $\mu$ m Critopoulos (1953), 61 - 90  $\mu$ m Raut and Ingle (1989).

On the basis of number of sclerotia per microscopic field the isolates were grouped into three groups as minimum (23 isolates), medium (11 isolates) and maximum (6 isolates) number of sclerotia. The number of sclerotia ranged from 8 to 23. In the present finding 57.5% of the isolates were having minimum number of sclerotia these results are confirmed with the findings of Shekhar *et al.* (2006) who reported that Hyderabad isolate produced highest number of sclerotia (180.3 sclerotia /9 mm disc and 52 / microscopic 10X field) whereas Coimbatore isolate produced minimum number of sclerotia (169 sclerotia).

Colour of sclerotia of all the isolates were varied and thus categorized as brown (6 isolates), dark brown (24 isolates) and black (10 isolates). The above

findings were also reported by Mandal *et al.* (1998). He reported light to dark brown sclerotia. Mandal *et al.* (1998) and Ratnoo *et al.* (1997) reported light to dark brown sclerotia in his investigation while Byadgi and Hegde (1985) reported brown to black colour sclerotia in his investigation.

Shape of sclerotia was also observed for each isolates and categorized into two groups as oval shape and round shape. Out of the 40 isolates 12 (30%) isolates were found as oval shape and 28 (70%) isolates were found as round shape of sclerotia. The above findings was also reported by Shekhar *et al.* (2006) they found that oblong shape having irregular edges and round shape with regular edges of sclerotia. Mandal *et al.* (1998) reported irregular and round to elongate sclerotial shape.

Pathological variability has been established by inoculating 5 days old seedling of BG-212 from all the 40 isolates of *R. bataticola* by using blotter paper technique. On the basis of variation exists in their virulence leading to the development of disease rating (1-9 scale), the isolates may be grouped into 2 categories viz; virulent (10 isolates), and 30 isolates as highly virulent. Highly virulent being the highest, as the maximum number of the isolates (75%) belongs to this category. Shekhar *et al.* (2006) reported variation in pathogenic behavior of *M. phaseolina*. and grouped as most virulent and least virulent isolates. Byadgi and Hegde (1985) studied and classified 7 isolates into three groups; A (Virulent) B (Intermediate) and C (Mild) on the basis of pathogenicity. Other workers have also reported the differential reaction by the isolates of *R. bataticola* in chickpea (Ratnoo *et al.* 1997; Singh *et al.* 1990; Sobti and Sharma 1992; Monga and Sheo Raj, 1994) and found considerable variation among isolates.

Present studies conducted on various aspects of variability in the isolates of *Rhizoctonia bataticola* led to be conclusion that the isolates pertaining to fast

growth rate (83.3%), partial submerged (90.1%), very thin hyphal width (88.8%), sclerotia bold size (83.3%), and dark brown (78.1%) and production of sclerotia (100%) is positively correlated with the virulence of as is reflected by the isolates number (1, 2, 3, 5, 6, 8, 9, 12, 13, 14, 16, 18, 20, 21, 22, 23, 24, 25, 27, 28, 29, 30, 31, 32, 33, 34, 35, 37, 38 and 39). The reaction produced by these isolates in susceptible variety (BG-212) is highly pathogenic giving disease rating of 8–9. Similar conclusion was drawn by Monga and Sheoraj (1994) in case of cottony root rot incited by *M. phaseolina*.

The available advanced breeding lines / entries were screened in the artificially infested multiple diseases sick plot. On the basis of percent mortality the lines were graded as resistant / moderately resistant / tolerant / susceptible / highly susceptible. Eleven entries namely BCP-21, BGD-130, BGD-9812, CSJ-140, CSJ-313, CSJ-437, CSJ-449, GL-22044, IPC-2000-25, MPJG-2000-25 and MPJG-2001-99-115 showed 0 to 10 per cent mortality and comes under resistant and 36, 13, 3 and 2 entries comes under moderately resistant, tolerant, susceptible and highly susceptible groups respectively.

A large number of cultivars of chickpea have been screened and identified resistant lines by various workers (Nene *et al.* 1989; Reddy *et al.* 1990; Bhatt 1993; Gangwar *et al.* 2002; Pandey *et al.* 2004; Gupta 1997; Gupta and Babbar, 2006; Pandey *et al.* 2004)

## SUMMARY, CONCLUSIONS AND SUGGESTIONS FOR FURTHER WORK

### 6.1 Summary

In Indian agriculture pulses play an important role not only from economical point of view but also due to their nutritional value. Several seed, soil and air borne diseases affect it, among the biotic stresses dry root rot caused by *Rhizoctonia bataticola* has been reported in many countries as a major yield-limiting factor. Forty samples of chickpea dry root rot affected plants were collected in 23 districts of Madhya Pradesh, Chhattisgarh, Maharashtra and Gujarat states under Wilt Network Project.

Diagnosis of symptoms of disease at different stage of crop growth revealed the disease to be dry root rot. The symptoms were mainly brittleness of taproot and presence of minute sclerotia and on isolation yielded *R. bataticola* (Taub.) Butler.

Variability among the isolates was further studied in respect to cultural morphological and pathological characterization. Cultural studies of all 40 isolates of *R. bataticola* revealed that isolates differ in the type of mycelial growth and growth rate on potato dextrose agar medium to be favorable for luxuriant growth for all the isolates.

Morphological studies of different isolates of *R. bataticola* revealed variability in size of hyphal width (5.2-9.3), size of sclerotia (72.7-117.2 X 57.1-106.5) number of sclerotia (8-23), on the basis of colour of sclerotial measurement the sclerotia were categorized into three group i.e., brown, dark brown and black; while looking to the shape of sclerotia, the isolates were categorized into two groups i.e., round and oblong.

Pathogenicity test was conducted by inoculating BG-212 a dry root rot susceptible variety of chickpea with 40 isolates by using blotter paper technique. All the isolates exhibited clear-cut variation in their virulence and characterized as, virulent and highly virulent. Indicating the possibility of two pathotypes and may be correlated them with the virulence of the isolates.

Studies on host resistance under multiple disease sick field led to conclusion that among all the entries, 11 entries namely BCP-21, BGD-130, BGD-9812, CSJ-140, CSJ-313, CSJ-437, CSJ-449, GL-22044, IPC-2000-25, MPJG-2000-25 and MPJG-2001-99-115 were found resistant to dry root rot, which can be utilized in breeding programmed for evolving high yielding dry root rot resistant acceptable cultivars of chickpea.

## 6.2 Conclusion

Chickpea (*Cicer arietinum* L.) is an important rabi pulse crop in Madhya Pradesh. Studies on 40 isolates of *R. bataticola* obtained from chickpea in different districts of Madhya Pradesh, Chhattisgarh, Maharashtra and Gujarat showed considerable variability in cultural, morphological and pathological characters and may be grouped into 2-3 pathotypes.

Present studies conducted on various aspects of variability in the isolates of *Rhizoctonia bataticola* led to be conclusion that the isolates having fast growth (83.3%), partial submerged (90.1%), very thin hypha (88.8%), sclerotia bold size (83.3%) and dark brown (78.1%) and production of sclerotia (100%) is positively correlated with the virulence.

Results of field screening for locating the sources of resistance revealed that, BCP-21, BGD-130, BGD-9812, CSJ-140, CSJ-313, CSJ-437, CSJ-449, GL-22044, IPC-2000-25, MPJG-2000-25 and MPJG-2001-99-115 showed resistant reaction.

### 6.3 Suggestions for further work

- Pathogenic variation in different isolates of *R. bataticola* by using differentials for known genotypes / varieties need to be further investigated under lab and field conditions.
- Variations in different isolates of *R. bataticola* on their molecular basis need to be worked out.
- There is need to identify stable sources of multiple disease resistance under field conditions.

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## VITA

The author of this thesis Mr. Surendra Singh Patel was born on 18<sup>th</sup> of Jan. 1984. He passed his high school examination from Govt.H.S.School Kondtari with 52.4% and higher secondary school examination from Govt. Multi. Natwar H.S. School Raigarh (C.G.) with 62.4% of marks.

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